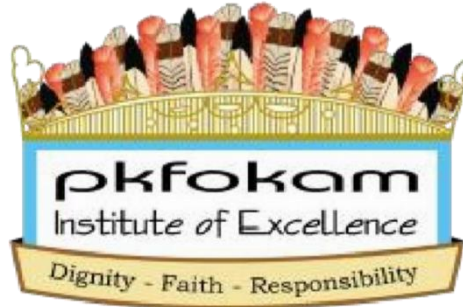




PKFOKAM INSTITUTE OF EXCELLENCE

PROJECT: WEDDING INVITATION AND GIFT MANAGEMENT SYSTEM

PROJECT REPORT

GIT HUB REPOSITORY AND PERSON MANAGEMENT CREATION

Course: C Programming

Group Members	Participation	Matricules
Matchim Dzuyim Verenne Celia (Group Leader)	22%	26S031
Djomo Yepdi Evans	19%	26S001
Talla Tchossi Daniella Elsa	21%	26S018
Mekolle Tuatsop Dalhia Aure	19%	26S004
Messi Zang Paul Jacques	19%	26S005

Lecturer: Mr. Mbietieu Amos Mbietieu

Academic Year: 2025/2026

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1. Project Overview

The **Wedding Invitation and Gift Management System (WIGMS)** is a software application designed to assist in organizing and managing wedding events. Weddings in Cameroon often involve a large number of guests, gift contributions, and multiple social categories, which makes manual management difficult.

The purpose of this system is to simplify wedding planning by providing tools to manage:

- Guest information
- Social categories of guests
- Invitation priority
- Wedding gift tracking

The application is developed using the **C programming language** and follows a **modular programming approach** where each part of the system is implemented as an independent module.

The system aims to improve organization, reduce errors, and help wedding planners efficiently manage guest lists and gifts.

Tool	Purpose
C Programming Language	Main programming language used for system implementation
GCC Compiler	Used to compile and run the C programs
Visual Studio Code / CodeBlocks	Code editor for writing and debugging the program
GitHub	Version control and collaboration platform
Standard C Libraries (stdio.h, string.h, stdlib.h)	Used for input/output operations, memory management and string handling

Table 1: Software tools and technologies used

2. Development Tools Used

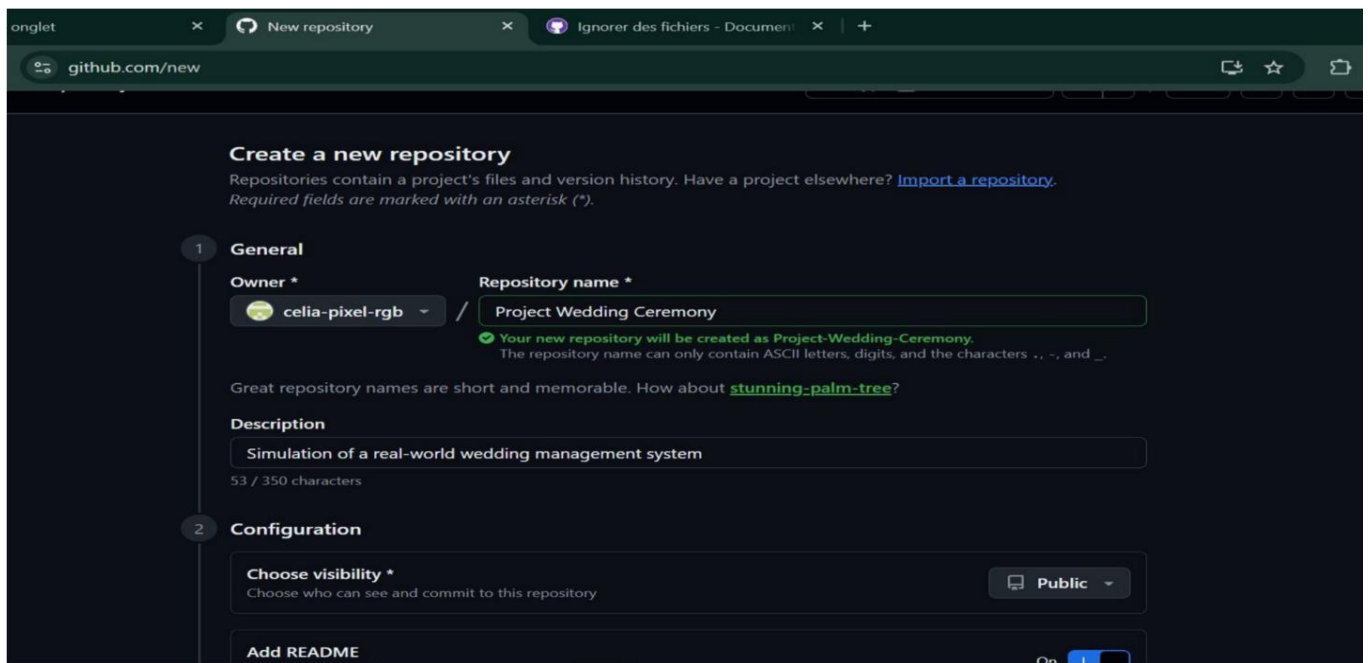
The following tools were used during the development of the system:

Table 2: GitHub repository configuration settings

SETTINGS	VALUE CHOSEN
Repository name	Project-Wedding-Ceremony
Visibility	Public
.gitignore	C template selected
Repository link	https://github.com/celia-pixelrgb/Project-Wedding-Ceremony

3. Github repository

3.1 Creating GitHub repository



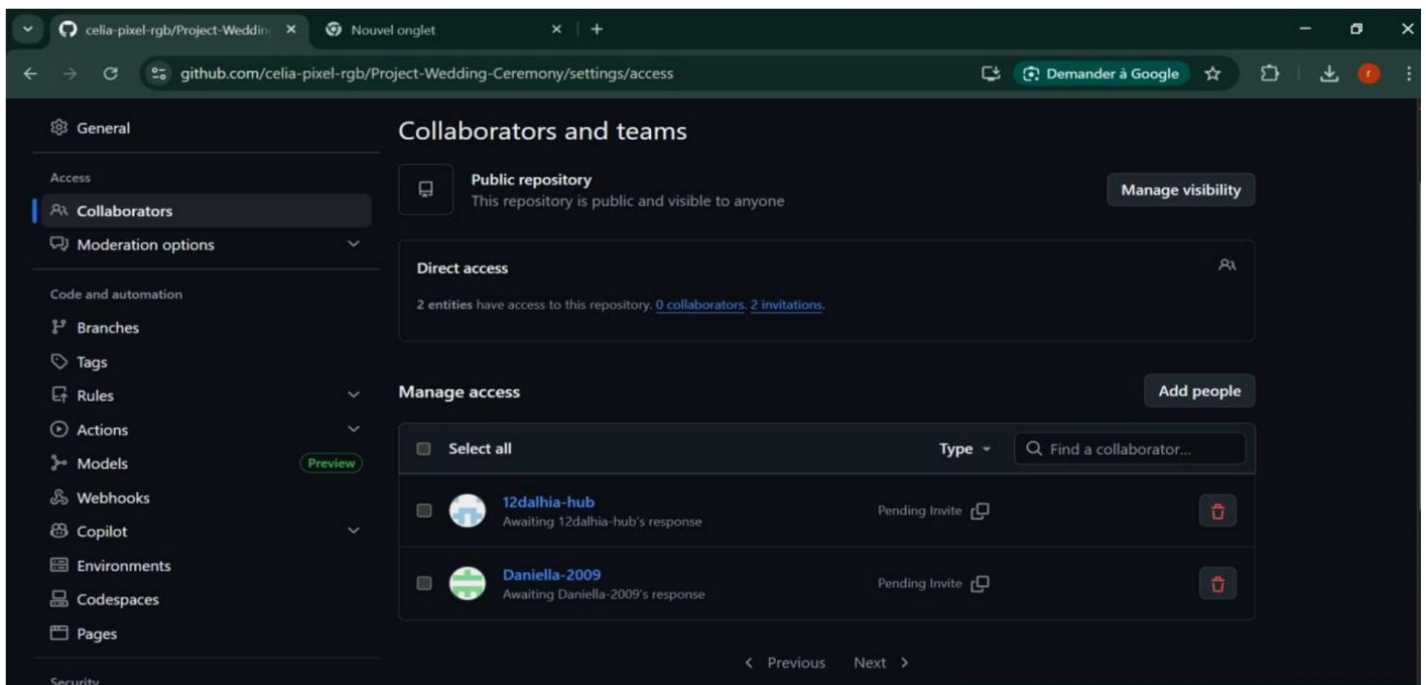
The group leader created the public repository with the following settings:

Here is the GitHub link to access our repository:

<https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

3.2. Adding Collaborators

The remaining group members were added as collaborators through the Setting --> collaborators page of the repository. Each member received an Email invitation from GitHub which they accepted in order to gain push access to the repository.



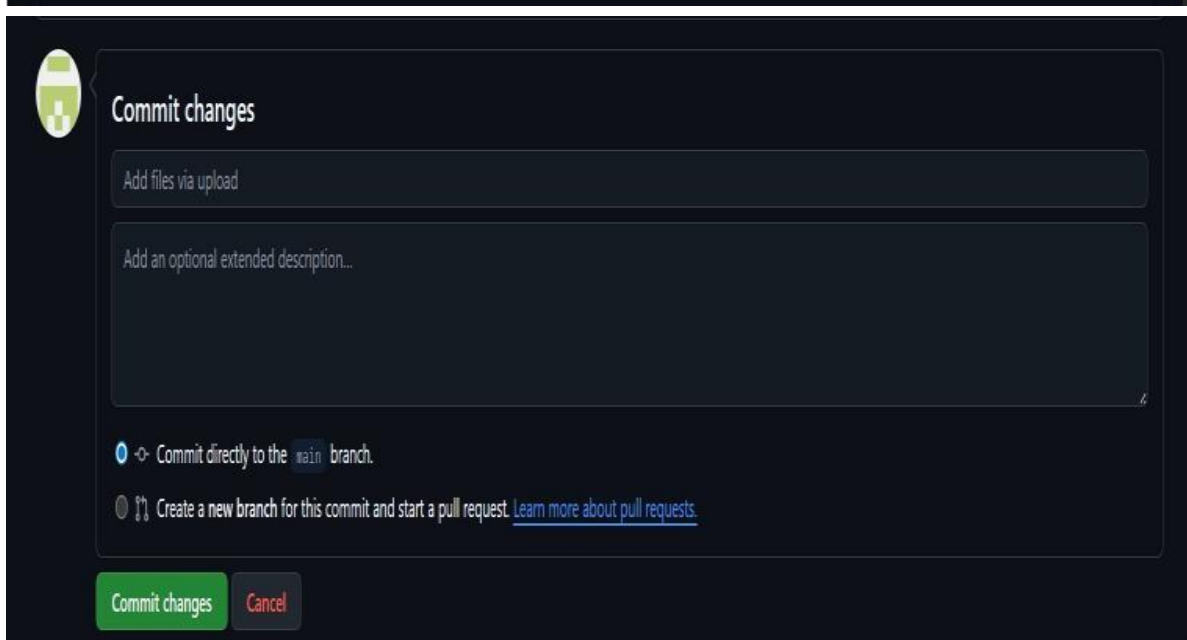
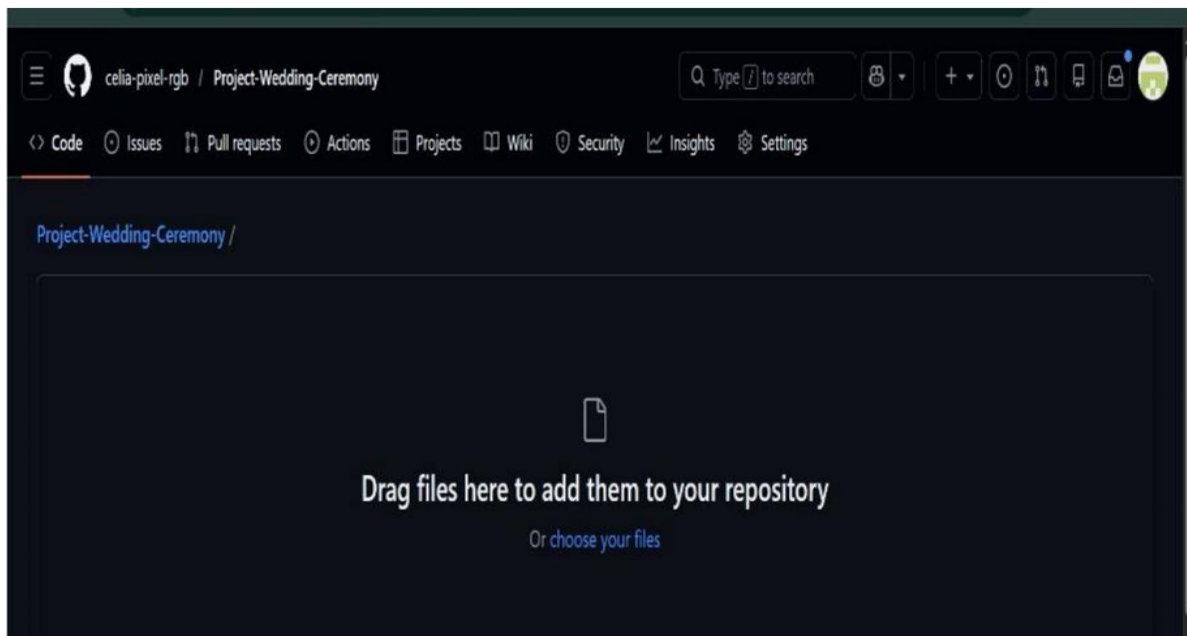
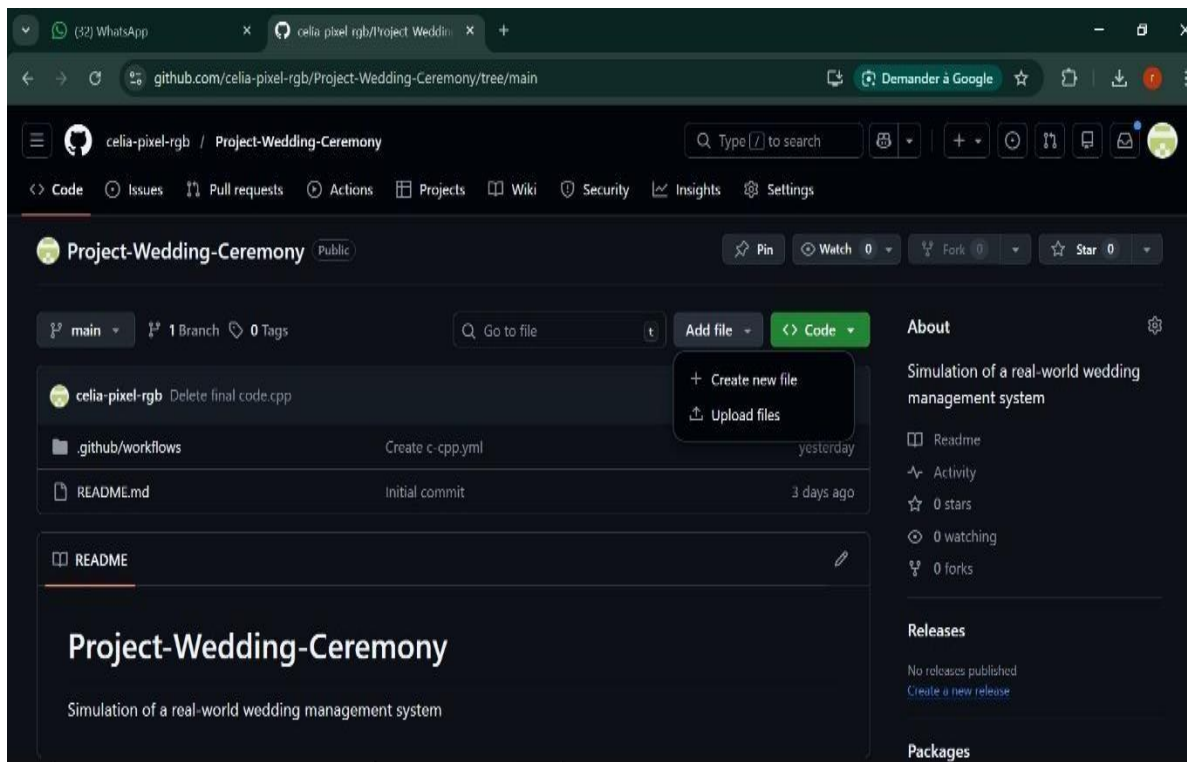
3.3. Cloning the Repository and first Push

After accepting their invitation, each member cloned the repository into their local machine. The group leader then performed the first push of the program.

```
Microsoft Windows [version 10.0.17763.1577]
(c) 2018 Microsoft Corporation. Tous droits réservés.

C:\Users\user>git clone https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony.git
Cloning into 'Project-Wedding-Ceremony'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

C:\Users\user>
```

4. Work Organisation and task distribution

The group consisted of 5 members. The work organisation depended on assigning responsibilities with respect to skills and the project phases. This approach permitted us to ensure that each section of this report would benefit from dedicated expertise.

Group Member	Main Role	Assigned Tasks
1. Matchim Dzuyim Verenne Celia	Implementation and testing	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: GTK, CSV, Test of Person Management, Test of Category Management.
2. Mekolle Tuatsop Dalhia Aure	Documentation and planning	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Report 1, Gantt Project, Report 2, Implementation of Category Management.
3. Talla Tchossi Daniella Elsa	Development (core modules)	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: GTK, CSV, Implementation of Person Management, implementation of category management Implementaton of gift management.
4. Messi Zang Paul Jacques	System design (UML diagrams)	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Use Case Diagram, Class Diagram.
5. Djomo Yepdi Evans	System modeling and deployment	Common tasks: Project Overview, Development Tools Used, System Architecture, Phase 1 Summary. Individual tasks: Sequence Diagram, Deployment Diagram.

Table 3: Work organisation and task distribution

5. System Architecture

The application follows a **simplified modular architecture inspired by the MVC design pattern**. The architecture consists of three main components:



Model

The model contains the core data structures and logic of the program, including:

- Person management
- Category management
- Priority management
- Gift management



Controller

The controller handles the flow of the application by managing the interactions between the user interface and the program modules.



View

The view represents the interface presented to the user. In this project, the view is implemented through a **console-based menu interface** that allows the user to navigate through the different system functionalities.

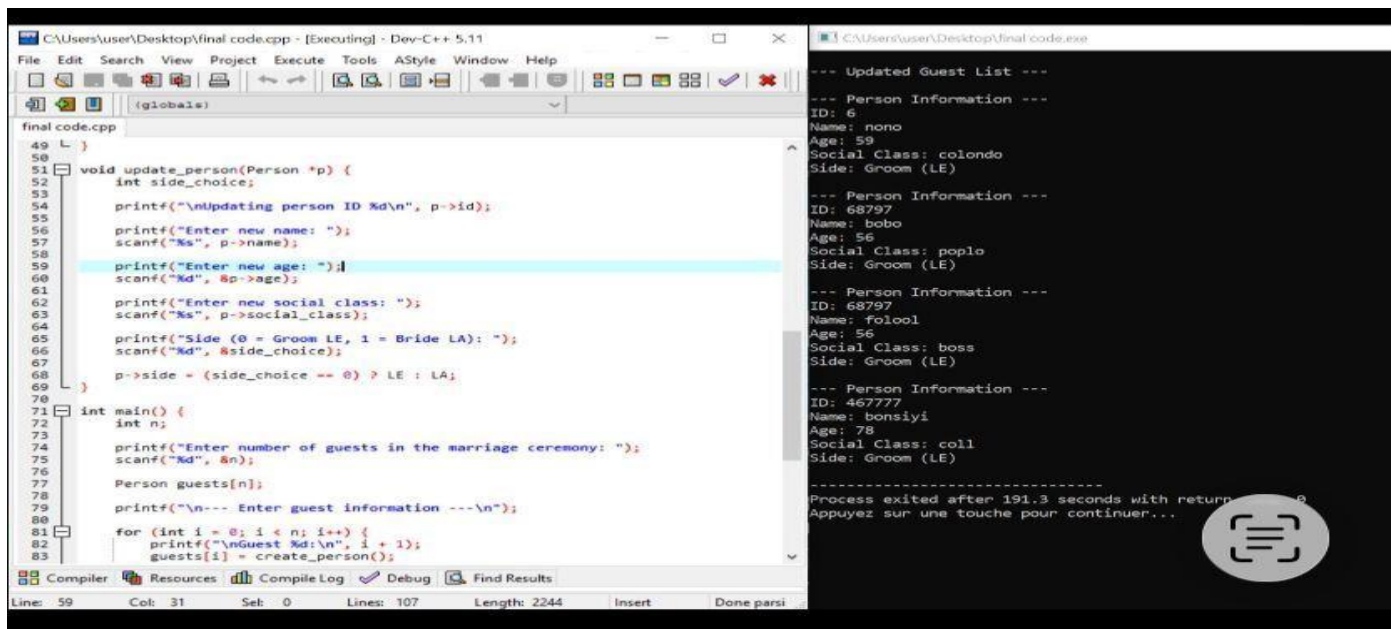
This architecture ensures clear separation between data, logic, and user interaction.

6. Functional Modules of the Application

As first stage, we did what we consider as our Module 1: Personal Management. This module focuses on:

6.1. Create a new guest

Here, we are creating a program that can create persons (guests). Based on the wanted total number of guests entered by the user, we defined a structure in which we register guests base on their name, age, Id, social status and their side (either the bride or the groom).



The screenshot shows a C++ IDE with two windows. The left window displays the source code for 'final code.cpp', and the right window shows the program's execution output.

```
final code.cpp
49 }
50
51 void update_person(Person *p) {
52     int side_choice;
53     printf("\nUpdating person ID %d\n", p->id);
54     printf("Enter new name: ");
55     scanf("%s", p->name);
56     printf("Enter new age: ");
57     scanf("%d", &p->age);
58     printf("Enter new social class: ");
59     scanf("%s", p->social_class);
60     printf("Side (0 = Groom LE, 1 = Bride LA): ");
61     scanf("%d", &side_choice);
62     p->side = (side_choice == 0) ? LE : LA;
63 }
64
65 int main() {
66     int n;
67     printf("Enter number of guests in the marriage ceremony: ");
68     scanf("%d", &n);
69     Person guests[n];
70     printf("\n--- Enter guest information ---\n");
71     for (int i = 0; i < n; i++) {
72         printf("\nGuest %d:\n", i + 1);
73         guests[i] = create_person();
74     }
75 }
```

The execution output on the right shows the program's runtime behavior, including prompts for guest information and the final list of guests.

```
--- Updated Guest List ---
--- Person Information ---
ID: 6
Name: nono
Age: 59
Social Class: colondo
Side: Groom (LE)
--- Person Information ---
ID: 68797
Name: bobo
Age: 56
Social Class: poplo
Side: Groom (LE)
--- Person Information ---
ID: 68797
Name: folool
Age: 56
Social Class: boss
Side: Groom (LE)
--- Person Information ---
ID: 467777
Name: bonsiyi
Age: 78
Social Class: coll
Side: Groom (LE)
-----
Process exited after 191.3 seconds with return value 0
Appuyez sur une touche pour continuer...
```

6.2. Display the guest

```
void display_person(Person p) {
    printf("\n--- Person Information ---\n");
    printf("ID: %d\n", p.id);
    printf("Name: %s\n", p.name);
    printf("Age: %d\n", p.age);
    printf("Social Class: %s\n", p.social_class);

    if (p.side == LE)
        printf("Side: Groom (LE)\n");
    else
        printf("Side: Bride (LA)\n");
}
```

7. Main Difficulties Encountered

During the development of the project, several challenges were encountered:

- Understanding the use of **structures in C programming**
- Correctly implementing **alternatives structures, loops and memory management**
- Designing linked list structures for dynamic data
- Managing user input validation
- Debugging errors during program execution

These challenges required careful testing and collaboration among team members.

8. Solutions Implemented

To address the difficulties encountered, the following solutions were implemented:

- Reviewing C programming documentation and tutorials
- Testing each function independently before integration
- Using debugging techniques to identify program errors
- Collaborating through GitHub to track and fix problems

These solutions allowed the team to successfully implement the first module of the project.

9. Phase 1 Summary

The first phase of the project was successfully completed.

The team was able to:

- Understand the project requirements
- Set up the development environment
- Create the project structure
- Implement the **Person Management module**
- Test the system functionality

The project is now ready to proceed to the next phase, which will involve implementing additional modules that we will present later on.

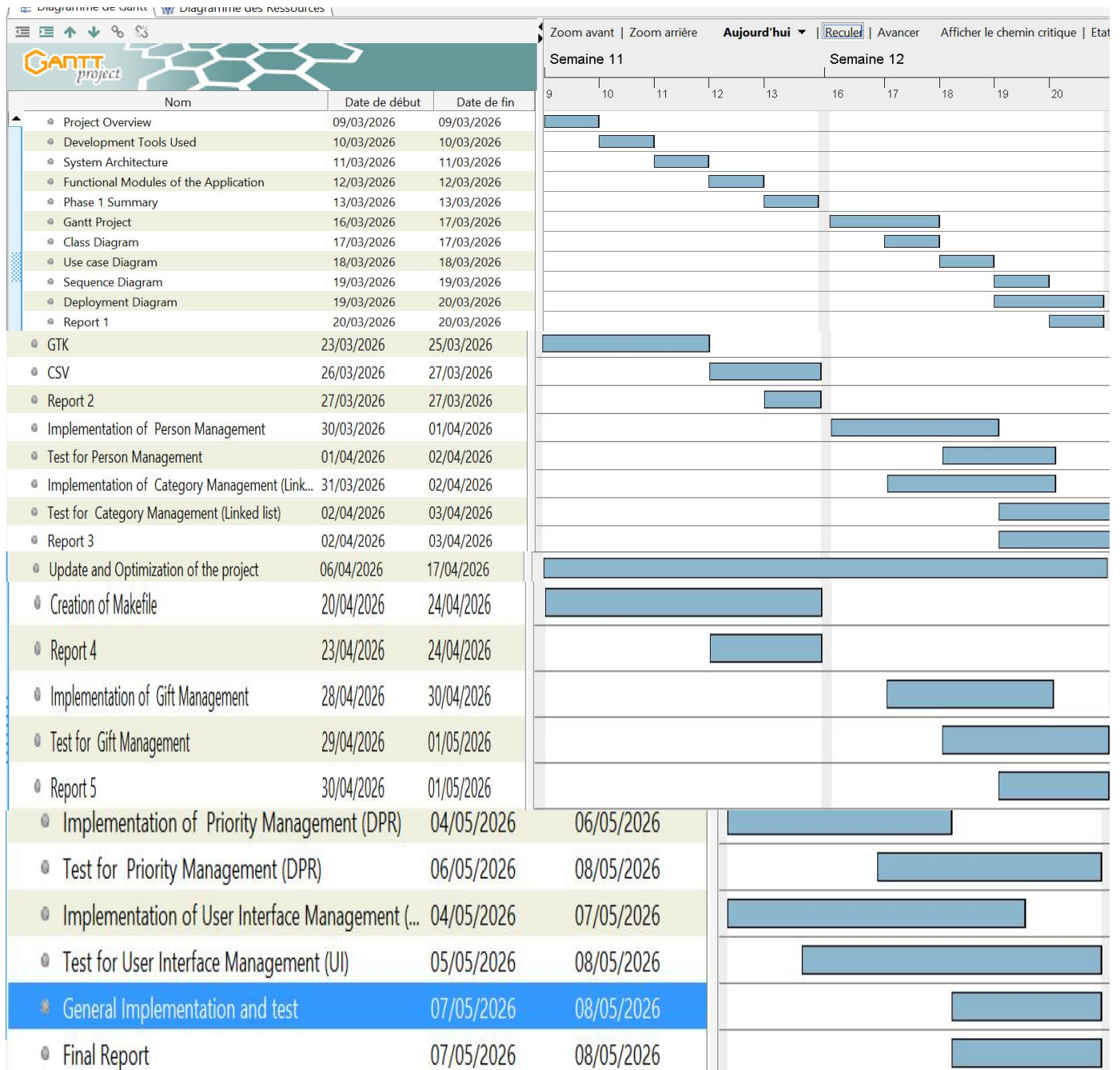
10. References

The following references were used during the development of this project:

- **Brian W. Kernighan & Dennis M. Ritchie** – *The C Programming Language*
- **Stephen Prata** – *C Programming: A Modern Approach*
- **ISO C Programming Documentation**
- **GitHub Official Documentation** – <https://docs.github.com>
- Course materials provided by the lecturer

11. Gantt Project

This project outlines the planning, design, and implementation of a software system using structured project management techniques. The Gantt chart was used to organize tasks, timelines, and dependencies efficiently.



The project was successfully structured using a Gantt chart, ensuring proper time management and task coordination. All phases were completed systematically, resulting in a functional and well-tested application.

12. SYSTEM DESIGN

System design is a critical phase in software development that defines the structure, components, and interactions of a system. It provides a clear blueprint to guide the development process and ensure that all functional and technical requirements are met.

In this section, we will present and explain the key design diagrams used in the project, including the class diagram, sequence diagram, deployment diagram, and use case diagram, to illustrate how the system operates and how its components interact.

12.1 CLASS DIAGRAM

The WIGMS UML Class Diagram models wedding organization entities using StarUML. A central Wedding class connects nine supporting classes — each with detailed attributes and defined relationships via UML associations and multiplicity notation.

- A class diagram provides a static view of the system, showing classes, attributes, operations, and relationships.
- In WIGMS, it acts as a blueprint for implementation.
- There are four classes; person, parking, gift and category

a. Class Diagram

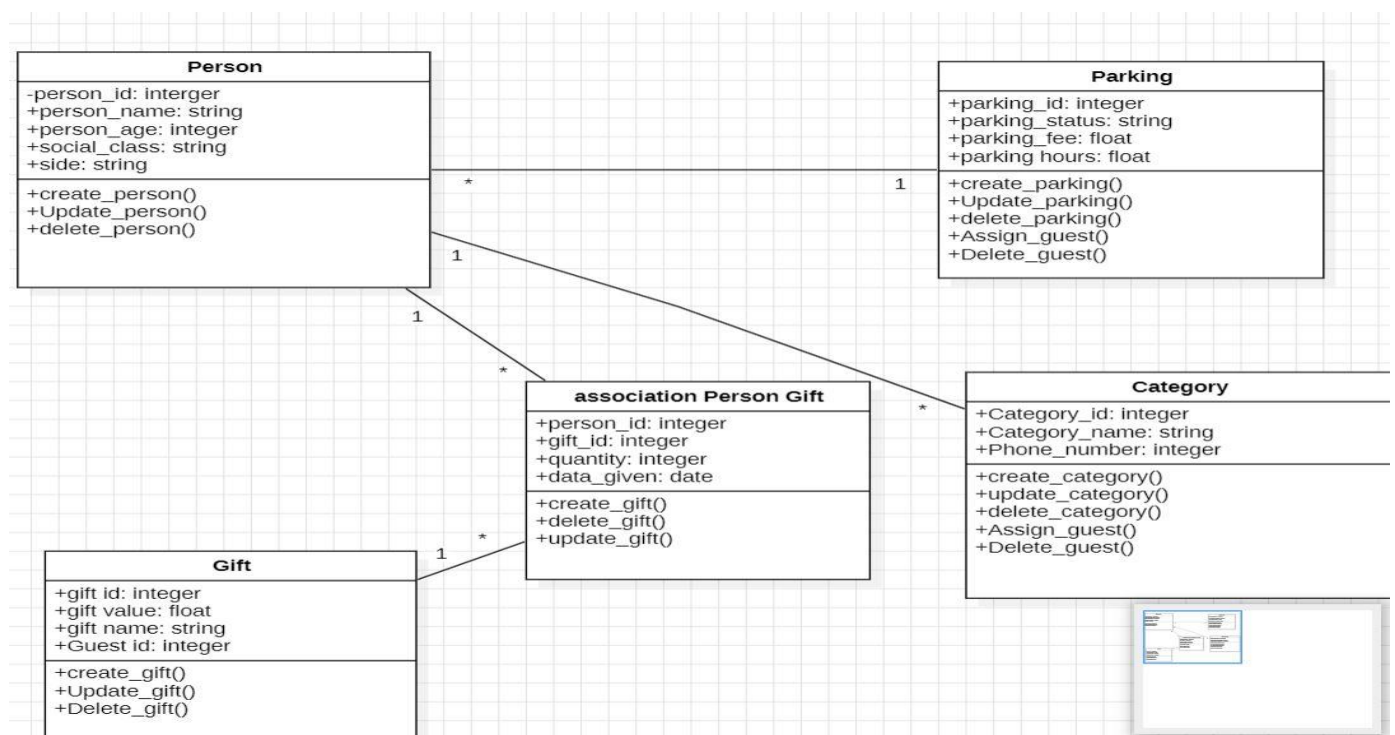
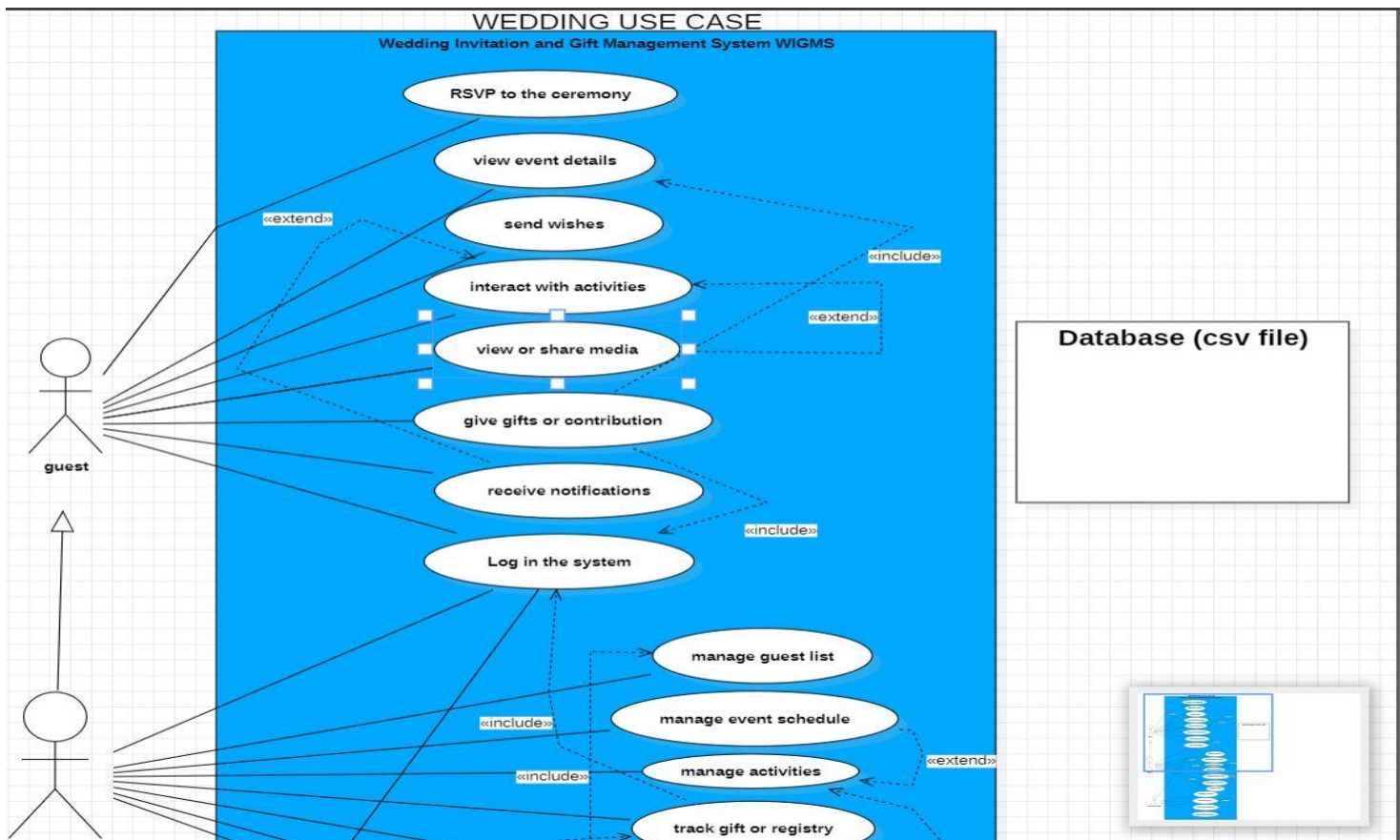


Figure 1: Class Diagram — Wedding Management System (StarUML)

Class	Attributes (private -)	Operations (public +)	Relationship
Person	id, name, age, social_class, side	create, update, delete	* → 1 Parking* → 1 Category* ↔ * Gift
Parking	id, status, fee	create, update	1 → * Person
Gift	id, value, name, guest_id	create, update, delete	* → 1 Category* ↔ * Person
Category	id, name, phone_number	create, update, delete	1 → * Person1 → * Gift

Table 4: System classes, attributes, and relationships

12.2 USE CASE DIAGRAM



12.3 SEQUENCE DIAGRAM

Overview

A sequence diagram illustrates how different components of the wedding-planning system interact over time through ordered message exchanges.

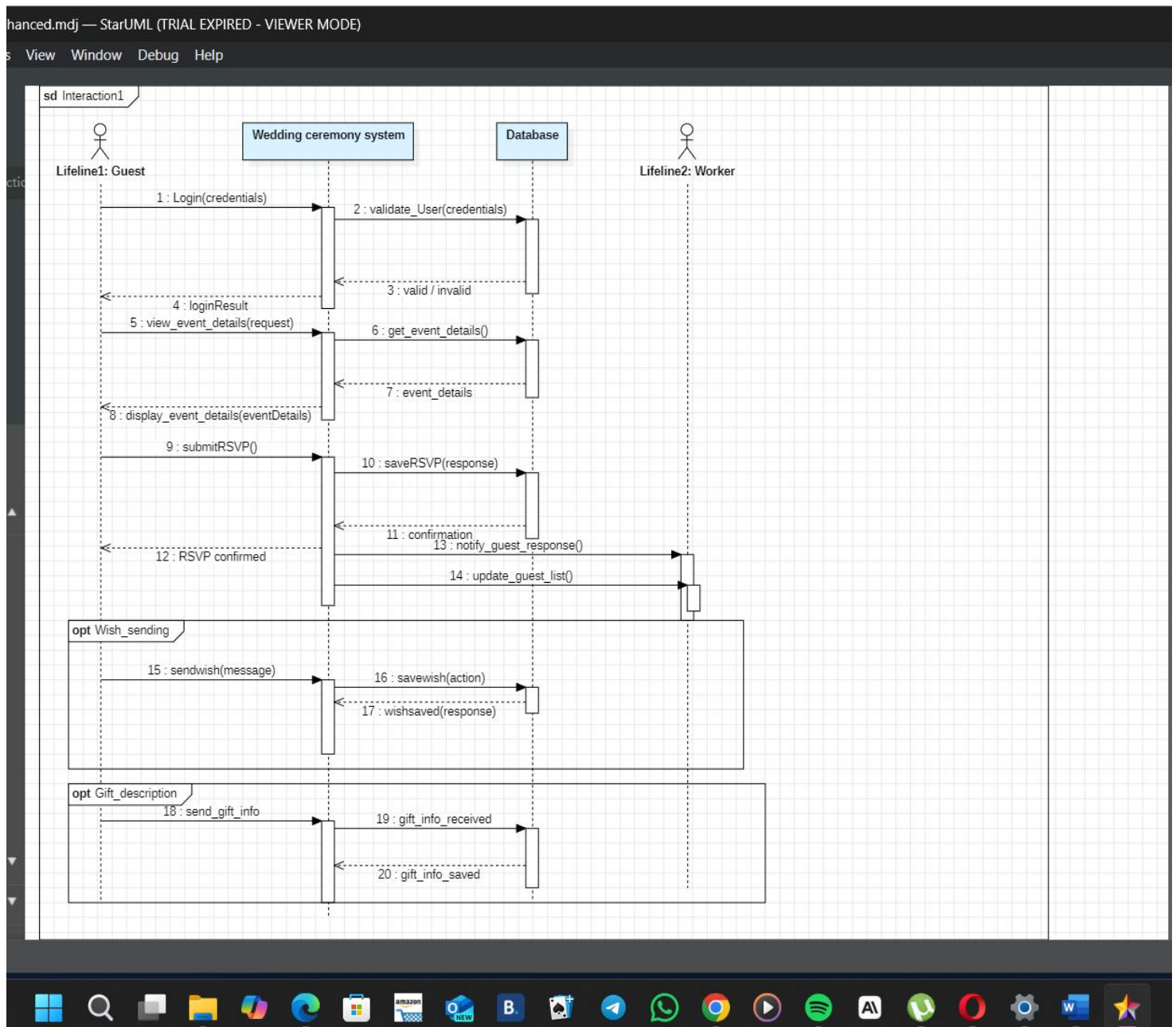


Figure 3: Sequence Diagram — Wedding Management System (StarUM)

It helps visualize the workflow of actions such as event creation, guest invitations, and RSVP tracking, ensuring processes occur in the correct sequence.

12.4 MNEMONIC TABLE

This table will enable any person familiar with c-programming projects who may or may have not participated in the project to better understand the actions going on represented on both the use case and sequence diagrams already illustrated above.

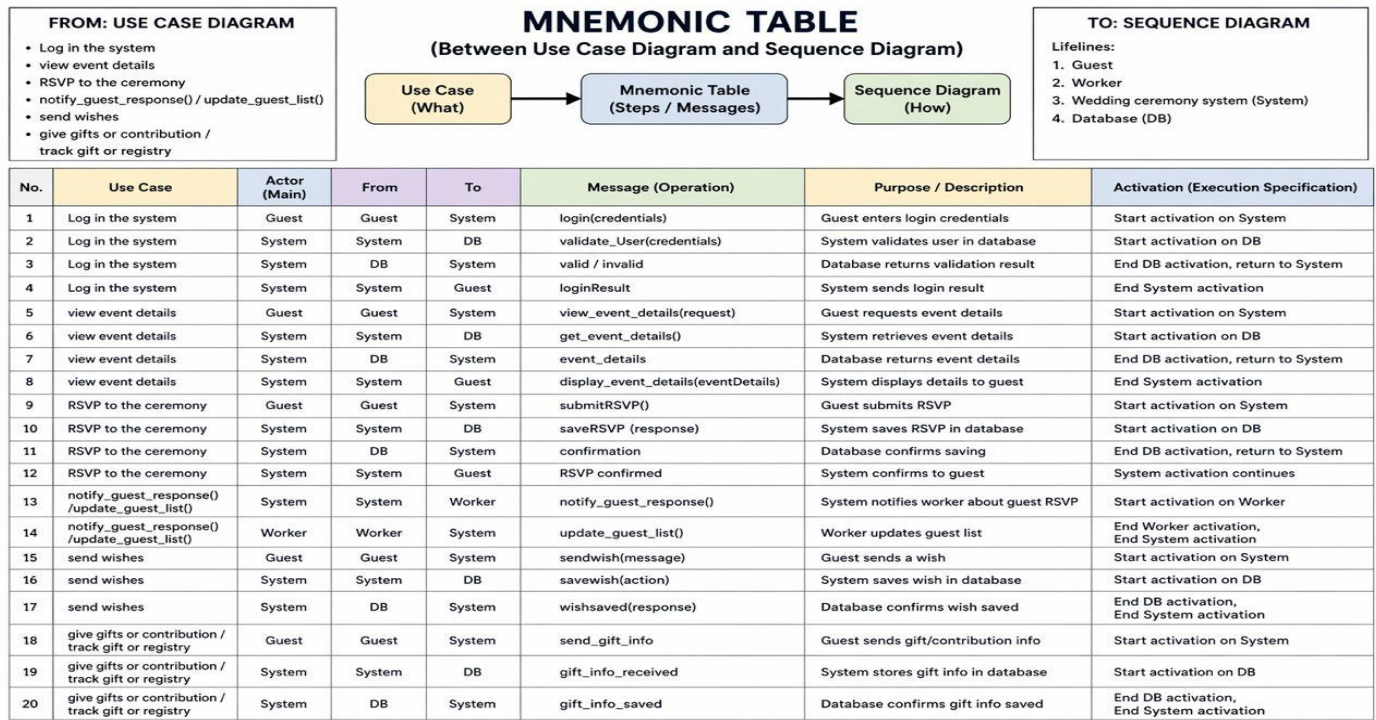


Figure 5: Mnemonic table between the usecase diagram and sequence diagram — Wedding Management System (StarUML)

12.5 DEPLOYMENT DIAGRAM

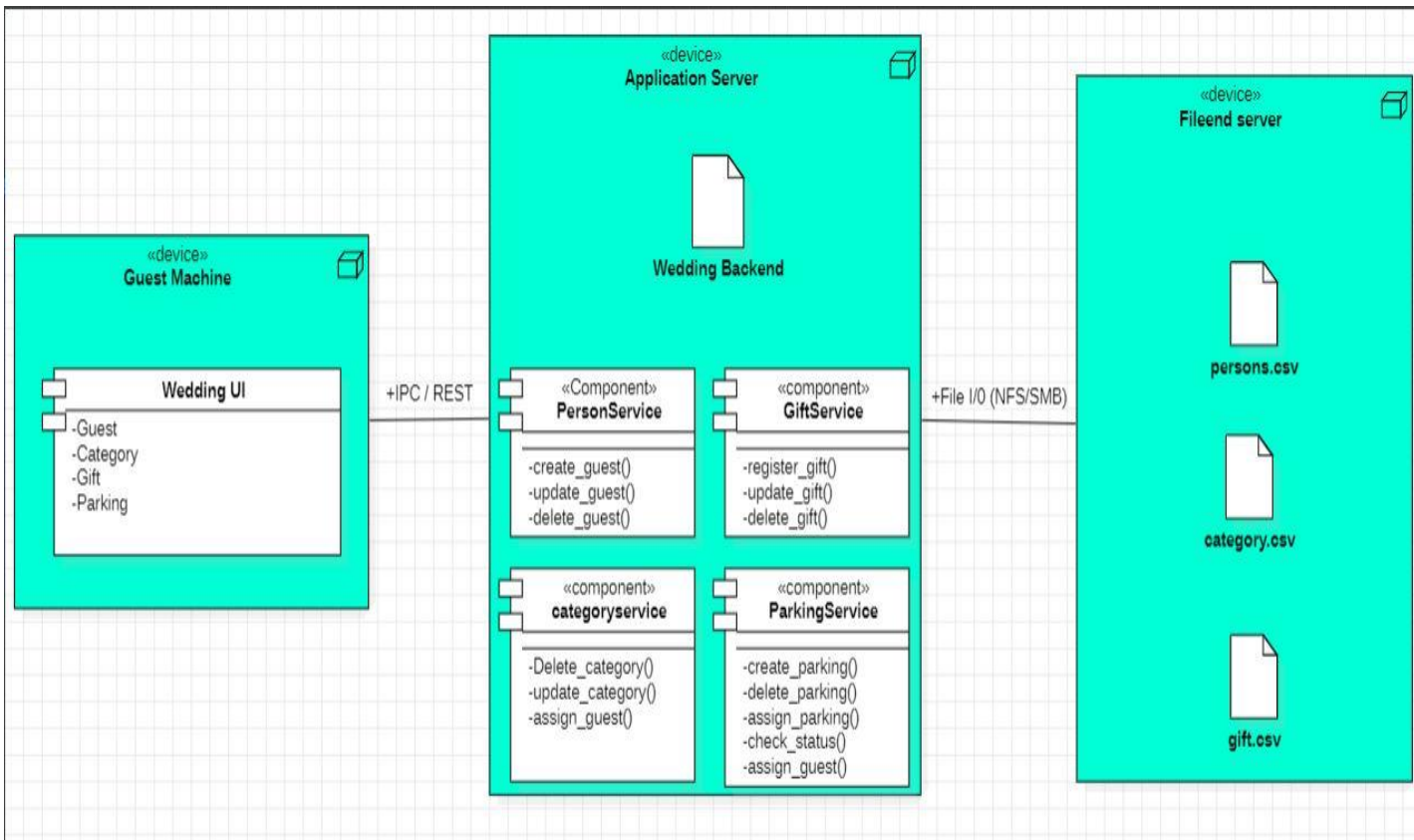


Figure 4: Deployment Diagram — Wedding Management System (StarUML)

13. GNU Image Manipulation Program Toolkit (GTK) and Comma-Selected Value (CSV)

13.1 GTK

Development of a GTK-based form application written in C for managing wedding guest data. The application collects user input through a graphical interface and stores it in a CSV file.

Program Overview

- **Download and Installation of GTK 4 for windows systems**

The program uses GTK4 to create a graphical interface consisting of input fields for Name, Age, Status,

First, you have to download the installer at msys2.org, selecting the first version for Windows.

1. Download the installer:

 [msys2-x86_64-20260322.exe](#)

or

 [msys2-arm64-20260322.exe](#)

(ARM64)

(Optional) For more information on the installer, like command line options, or how to verify the checksum and signature of the installer, see the [installer guide](#).

- Following this, at the end of the download process, ensure that you store the installer in a folder you will easily remember e.g wedding management ceremony.
- Again, open the “MSYS2 MSYS” terminal and run the command “pacman -Syu”, and press “Y” until you finalize the installations.

```
user@DESKTOP-S24HAFG MSYS ~
$ pacman -Syu
:: Synchronizing package databases...
clangarm64           521.3 KiB   12.4 KiB/s   00:42 [#####] 100%
mingw32             110.7 KiB    7.34 KiB/s   00:15 [#####] 100%
mingw64             459.4 KiB   21.8 KiB/s   00:21 [#####] 100%
ucrt64              538.8 KiB   13.1 KiB/s   00:41 [#####] 100%
clang64             529.2 KiB    9.69 KiB/s   00:55 [#####] 100%
msys                456.2 KiB   52.0 KiB/s   00:09 [#####] 100%
```

- Still, run the command “\$ pacman -S mingw-w64-ucrt-x86_64-gcc” into the “MSYS2 UCR64” terminal, and validate all the options till the end of the various installations.

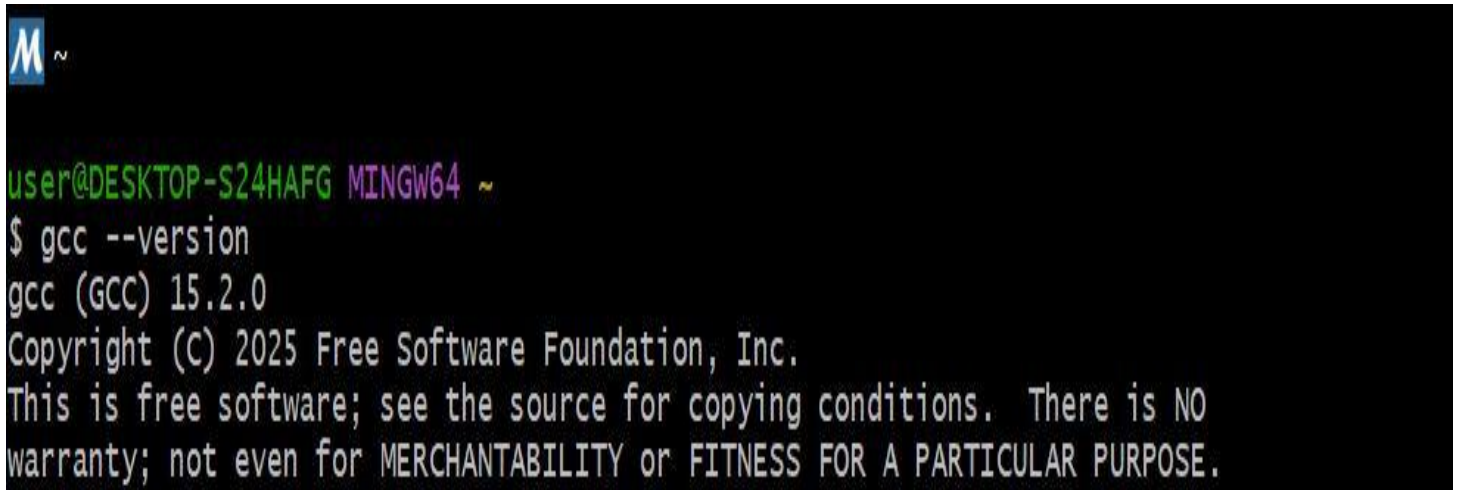
```
user@DESKTOP-S24HAFG UCRT64 ~
$ pacman -S mingw-w64-ucrt-x86_64-gcc
resolving dependencies...
looking for conflicting packages...

Packages (2) mingw-w64-ucrt-x86_64-gcc-libs-15.2.0-14  mingw-w64-ucrt-x86_64-gcc-15.2.0-14

Total Download Size:    47.67 MiB
Total Installed Size:  248.34 MiB
Net Upgrade Size:       0.07 MiB

:: Proceed with installation? [Y/n]
:: Retrieving packages...
mingw-w64-ucrt-x86_64-gcc-...  546.8 KiB   23.4 KiB/s   33:34 [-----] 1%
mingw-w64-ucrt-x86_64-gcc-...  804.7 KiB   40.2 KiB/s   00:06 [#####] 76%
Total (0/2)                  1351.5 KiB   73.2 KiB/s   10:48 [-----] 2%
```

- To continue, install GTK still in the “MSYS2 MINGW64” terminal by typing the command “pacman -S mingw-w64-x86_64-gtk4”, and then install build tools by typing-in the command “pacman -S mingw-w64-x86_64-pkgconfig mingw-w64-x86_64-make” in the same terminal.
- Next, in order to ensure that the gcc compiler was well installed, run the command “gcc --version” into the “MSYS2 MINGW64” terminal. You should see the version afterwards.

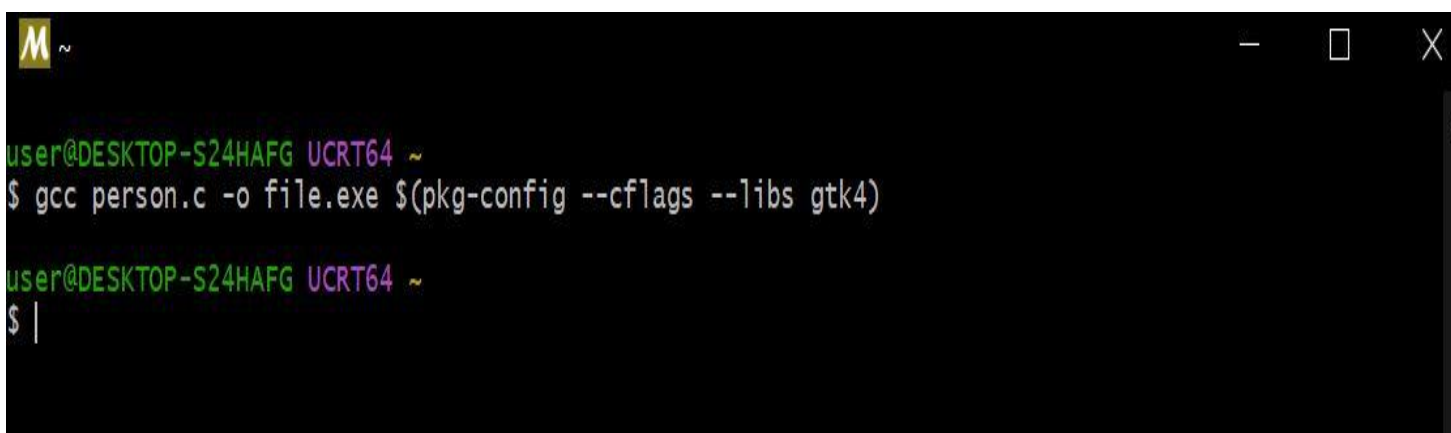
A terminal window with a dark background and a blue 'M' icon in the top left corner. The prompt is 'user@DESKTOP-S24HAFG MINGW64 ~'. The user has entered the command '\$ gcc --version'. The output is 'gcc (GCC) 15.2.0' followed by copyright information for the Free Software Foundation, Inc. and a disclaimer about warranty.

```
user@DESKTOP-S24HAFG MINGW64 ~  
$ gcc --version  
gcc (GCC) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

- You have now installed your GTK library, you should proceed further with the execution of codes.

NB

- ✓ Before running a code, you must make sure that it is stored in the same folder as the one containing your GTK library files.
- ✓ If the code ran successfully, a “.exe” file appears just below your c-code in the folder.
 - In order to run the code, you must open the “MSYS2 UCR64” terminal and type the command “gcc file.c -o file.exe \$(pkg-config

A terminal window with a dark background and a yellow 'M' icon in the top left corner. The prompt is 'user@DESKTOP-S24HAFG UCRT64 ~'. The user has entered the command '\$ gcc person.c -o file.exe \$(pkg-config --cflags --libs gtk4)'. The prompt is now '\$ |' on a new line.

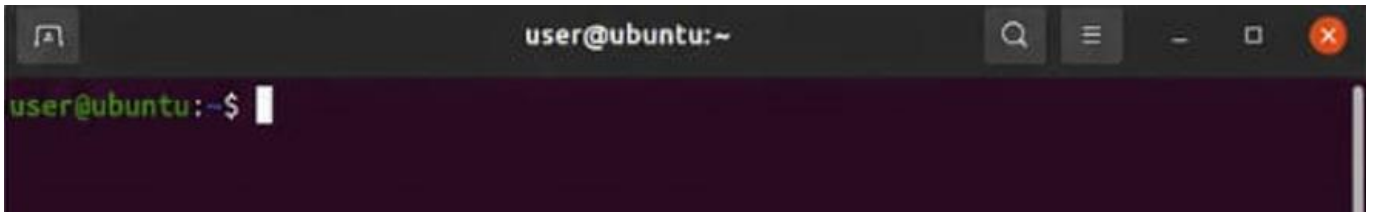
```
user@DESKTOP-S24HAFG UCRT64 ~  
$ gcc person.c -o file.exe $(pkg-config --cflags --libs gtk4)  
user@DESKTOP-S24HAFG UCRT64 ~  
$ |
```

–cflags –libs gtk4)”

- Download and installation of GTK for Linux

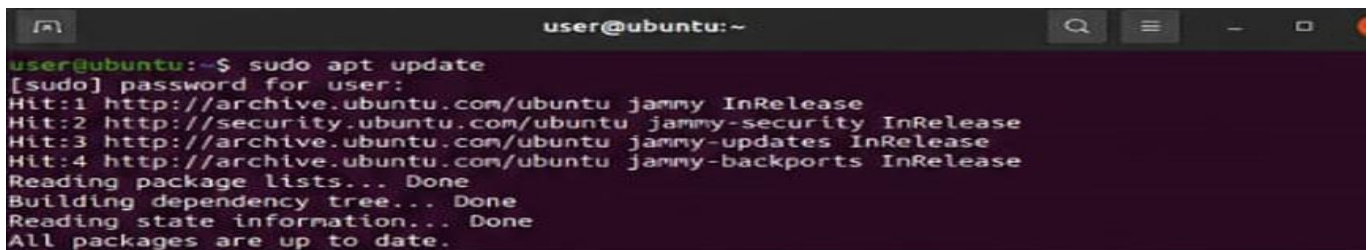
To install GTK for some particular systems as “Linux”, we follow different steps:

➤ Step1: Open the terminal



➤ Step2: Update your system

Here, you run the following command “*sudo apt update*” to update the package list.



➤ Step3: install GTK

You run the following command “*sudo apt install light-4-dev*” in order to install the GTK4 development packages



- After entering the command, you need to confirm the installation by “typing Y and press Enter”. Therefore, the packages will be downloaded and installed.

```
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgtk-4-1 amd64 4.10.2-1ubuntu1 [2,598 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 gir1.2-gtk-4.0 amd64 4.10.2-1ubuntu1 [107 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgtk-4-dev amd64 4.10.2-1ubuntu1 [4,110 kB]
Fetched 6,815 kB in 2s (3,329 kB/s)
Selecting previously unselected package libgtk-4-1:amd64.
(Reading database ... 197334 files and directories currently installed.)
Preparing to unpack .../libgtk-4-1_4.10.2-1ubuntu1_amd64.deb ...
Unpacking libgtk-4-1:amd64 (4.10.2-1ubuntu1) ...
Selecting previously unselected package gir1.2-gtk-4.0:amd64.
Preparing to unpack .../gir1.2-gtk-4.0_4.10.2-1ubuntu1_amd64.deb ...
Unpacking gir1.2-gtk-4.0:amd64 (4.10.2-1ubuntu1) ...
Selecting previously unselected package libgtk-4-dev:amd64.
Preparing to unpack .../libgtk-4-dev_4.10.2-1ubuntu1_amd64.deb ...
Unpacking libgtk-4-dev:amd64 (4.10.2-1ubuntu1) ...
Setting up libgtk-4-1:amd64 (4.10.2-1ubuntu1) ...
Setting up gir1.2-gtk-4.0:amd64 (4.10.2-1ubuntu1) ...
Setting up libgtk-4-dev:amd64 (4.10.2-1ubuntu1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.1) ...
user@ubuntu:~$
```

➤ Step4: Verifying the installation

After installing, you need to check the installed GTK4 version. To do it, you enter the command “*pkg-config --modversion gtk4*”

```
user@ubuntu:~$ pkg-config --modversion gtk4
4.10.2
user@ubuntu:~$
```

NB: In order to verify the installed GTK4 and its functionality, you can decide to create and test a program. To compile the program, you need to enter the following command

“*gcc name_of_the_program.c `pkg-config --cflags --libs gtk4` -o name_of_the_program*”

Then to test, we run: “*./name_of_the_program*”

EX:

```
user@ubuntu:~$ gcc test.c `pkg-config --cflags --libs gtk4` -o test
user@ubuntu:~$ ./test
user@ubuntu:~$
```


14. Key Implementation Details

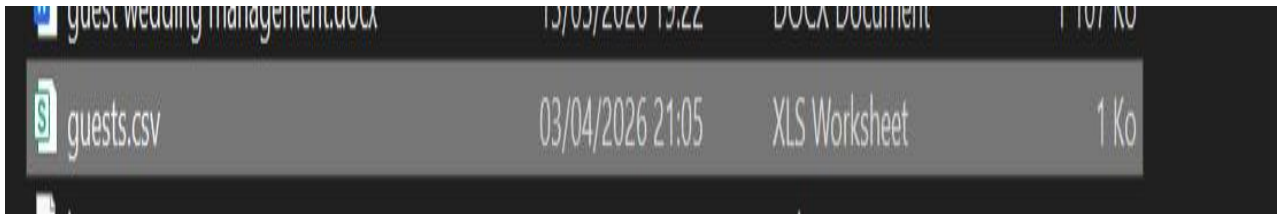
```
marriage.c  x
1  #include <gtk/gtk.h>
2  #include <stdio.h>
3
4  // Input fields (global so we can access them easily)
5  GtkWidget *name_entry;
6  GtkWidget *age_entry;
7  GtkWidget *status_entry;
8  GtkWidget *phone_entry;
9  GtkWidget *side_entry;
10
11 // Function called when button is clicked
12 void save_data(GtkButton *button, gpointer user_data) {
13
14     // Get text from input fields
15     const char *name = gtk_editable_get_text(GTK_EDITABLE(name_entry));
16     const char *age = gtk_editable_get_text(GTK_EDITABLE(age_entry));
17     const char *status = gtk_editable_get_text(GTK_EDITABLE(status_entry));
18     const char *phone = gtk_editable_get_text(GTK_EDITABLE(phone_entry));
19     const char *side = gtk_editable_get_text(GTK_EDITABLE(side_entry));
20
21     // Open CSV file
22     FILE *file = fopen("guests.csv", "a");
23
24     if (file == NULL) {
25         g_print("Error opening file\n");
26         return;
27     }
28
29     // Write data into file
30     fprintf(file, "%s,%s,%s,%s,%s\n", name, age, status, phone, side);
31
32     fclose(file);
33
34     g_print("Data saved!\n");
35 }
36
37 // Function that creates the window
38 void activate(GtkApplication *app, gpointer user_data) {
39     GtkWidget *window;
40
41     // Function that creates the window
42     void activate(GtkApplication *app, gpointer user_data) {
43
44         GtkWidget *window;
45         GtkWidget *grid;
46         GtkWidget *button;
47
48         // Create window
49         window = gtk_application_window_new(app);
50         gtk_window_set_title(GTK_WINDOW(window), "Wedding Form");
51         gtk_window_set_default_size(GTK_WINDOW(window), 400, 300);
52
53         // Create layout
54         grid = gtk_grid_new();
55         gtk_window_set_child(GTK_WINDOW(window), grid);
56
57         // Create input fields
58         name_entry = gtk_entry_new();
59         age_entry = gtk_entry_new();
60         status_entry = gtk_entry_new();
61         phone_entry = gtk_entry_new();
62         side_entry = gtk_entry_new();
63
64         // Add Labels and fields
65         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Name"), 0, 0, 1, 1);
66         gtk_grid_attach(GTK_GRID(grid), name_entry, 1, 0, 1, 1);
67
68         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Age"), 0, 1, 1, 1);
69         gtk_grid_attach(GTK_GRID(grid), age_entry, 1, 1, 1, 1);
70
71         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Status"), 0, 2, 1, 1);
72         gtk_grid_attach(GTK_GRID(grid), status_entry, 1, 2, 1, 1);
73
74         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Phone"), 0, 3, 1, 1);
75         gtk_grid_attach(GTK_GRID(grid), phone_entry, 1, 3, 1, 1);
76
77         gtk_grid_attach(GTK_GRID(grid), gtk_label_new("Side"), 0, 4, 1, 1);
78         gtk_grid_attach(GTK_GRID(grid), side_entry, 1, 4, 1, 1);
79
80         // Create button
81         button = gtk_button_new_with_label("Save");
82         gtk_grid_attach(GTK_GRID(grid), button, 0, 5, 2, 1);
83
84         // Connect button click to function
85         g_signal_connect(button, "clicked", G_CALLBACK(save_data), NULL);
86
87         gtk_window_present(GTK_WINDOW(window));
88     }
89
90     // Main function
91     int main(int argc, char **argv) {
92         GtkApplication *app;
93
94         app = gtk_application_new("org.example.wedding", G_APPLICATION_DEFAULT_FLAGS);
95         g_signal_connect(app, "activate", G_CALLBACK(activate), NULL);
96
97         return g_application_run(G_APPLICATION(app), argc, argv);
98     }
99 }
```

- ✓ **Global widget pointers** were used to allow access to input fields across functions. This simplifies retrieving user input when the Save button is clicked.
- ✓ **The use of `gtk_editable_get_text()` allows** extraction of user input from each entry field. This is required because GTK widgets do not directly expose their content.
- ✓ **File handling** is implemented using `fopen` in append mode ("a").
This ensures that new data is added without overwriting existing entries.

14.2 Csv file creation

The csv file was generated immediately after the GTK form worked, and was tested by inputting arbitrary entries

Person csv file



A	B	C	D
0,Matchim,16,VIP,679611233,0,Yes			

14.3 Problem Encountered and Solution

General Problem

- Difficulty in configuring development tools such as GTK.
- Limited understanding of UML diagrams (sequence, deployment, use case, class).
- Challenges in using StarUML (navigation, relationships, layout).
- Errors in coding such as file handling, compilation, and signal connections.
- Complexity in organizing system interactions and component communication.

14.4 General Solution

- Proper installation and configuration of development tools.
- Studying UML concepts and applying correct diagram rules.
- Practicing and exploring StarUML features for better usage.
- Applying debugging techniques and correct coding practices.
- Breaking the project into smaller tasks using structured planning tools like Gantt charts.

14.5 Conclusion

The Wedding Ceremony Management System demonstrates how C programming can be combined with project planning tools, system design diagrams, and user interface technologies. The use of GTK and CSV enhances usability and data management, making the system efficient and practical for real-world applications.

15. Person and category management: implementations & tests

15.1 Person Management

This module concerns the registration of the guests at a wedding ceremony, alongside all the operations carried out on them (Entry, Update, Delete).

- The previous model in the second report being lacking, the following is an updated version
- The c-program was written and thoroughly debugged using dev c++ software
- The previous program being too boring, because it was obliging the guest to restart back his entire registration when an error was committed during its registration, the following version is an updated version

15.1.1 Extract of Person Management C-code

```
person.c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include "person.h"
5
6  // this is used to represent which side the guest b
7  // from the person.h, we will have structure or par
8  // these are main storage file and temporary filed.
9  // ALL platforms (person.c, gtkperson.c) share this
10 const char *CSV_FILE = "persons.csv";
11 // this is to insure the security and the confident
12 const char *PASSWORD = "group3wed!";
13
14 //it converts enum to readable string
15
16 // reads a line of text from the user after printin
17 static void read_line(const char *prompt, char *buf
18     printf("%s", prompt);
19     fflush(stdout); //to ensur
20     if (fgets(buf, size, stdin))
21         buf[strcspn(buf, "\n")] = '\0'; // to remo
22 }
23
24 // reads a single integer from the user input
25 static int read_int(const char *prompt) {
26     char buf[32];
27     read_line(prompt, buf, sizeof(buf));
28     return atoi(buf);
29 }
```

Compiler Resources Compile Log Debug Find Results

Line: 11 Col: 52 Sel: 0 Lines: 373 Length: 13963 Insert

On this program, we can register a guest, display all guest, delete a guest and update a guest. The particular of this code

```
Choice: 2
Password: group3wed!

| ID | Name           | Age | Status | Phone       | Side | Parking |
|----|-----|-----|-----|-----|-----|-----|
| 0  | Talla Daniella | 16  | VIP    | 674836353   | Bride | No,Tallad |
| 1  | Matchim Celia  | 18  | VIP    | 674839363   | Bride | No,dzuyim |
| 2  | Lesly Yimdjo   | 22  | Single | 697854938   | Groom | No,leslyy |
| 3  | Lesly Kémogne  | 18  | Single | 697854938   | Groom | Yes,celia |
| 4  | Kounga Junior  | 23  | Married | 675854043   | Bride | No,kounga |
| 5  | Michele Lindsay | 17  | Single | 678595444   | Groom | Yes       |
| 6  | Matchi         | 16  | Single | 675848943   | Groom | No        |
| 7  | Matchoing      | 17  | Single | 678998659   | Groom | Yes       |
| 8  | moni           | 15  | Single | 675857444   | Groom | Yes       |

Wedding ù Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice:
```


is that, when committing a registration or updating error, the program does not start back in the beginning. It only mentions the error, and allows you to correct it.

a) Person Registration

b) Person Display

```
Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice: 1
Full Name: Matchoing
Age: 17
Status (Single/Married/Divorced): Single
Phone (9 digits, start 6-9): 675849444
Side: 1=Groom 2=Bride 3=Both
Choice: 1
Parking? 1=Yes 2=No
Choice: 1
Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice: 1
Full Name: moni
Age: 15
Status (Single/Married/Divorced): Single
Phone (9 digits, start 6-9): 6543335fr
Invalid phone (must be 9 digits starting with 6, 7, 8 or 9). Please try again.
Phone (9 digits, start 6-9): 675857444
Side: 1=Groom 2=Bride 3=Both
Choice: 2
Parking? 1=Yes 2=No
Choice: 1
Wedding Guest Manager
1. Add guest
2. Display all guests
3. Delete guest
4. Update guest
0. Exit
Choice: 2
Password: group3wed!
```

15.1.2 Extract of person management GTK c-code



```
22 // ----- PASSWORD -----
23 const char *PASSWORD = "group3wed!";
24
25 // ----- FORM -----
26 GtkWidget *name_entry, *age_entry, *status_entry, *phone_entry;
27 GtkWidget *radio_groom, *radio_bride, *radio_both;
28 GtkWidget *parking_dropdown;
29
30 // ----- DISPLAY -----
31 GtkWidget *manager_text;
32 GtkWidget *display_password_entry;
33
34 // ----- UPDATE -----
35 GtkWidget *update_id_entry, *update_name, *update_age, *update_status, *update_phone;
36 GtkWidget *update_radio_groom, *update_radio_bride, *update_radio_both;
37 GtkWidget *update_parking;
38 GtkWidget *update_password_entry;
39 GtkWidget *update_fields_box;
40
41
```

-
- The csv file was automatically generated below the gtk person management file immediately the gtk compiled successfully.

15.1.3 GTK ran successfully

```
M ~
USER@DESKTOP-KSHPOJE UCRT64 ~
$ gcc person.h person.c -o person.exe $(pkg-config --cflags --libs gtk4)
USER@DESKTOP-KSHPOJE UCRT64 ~
$ |
```

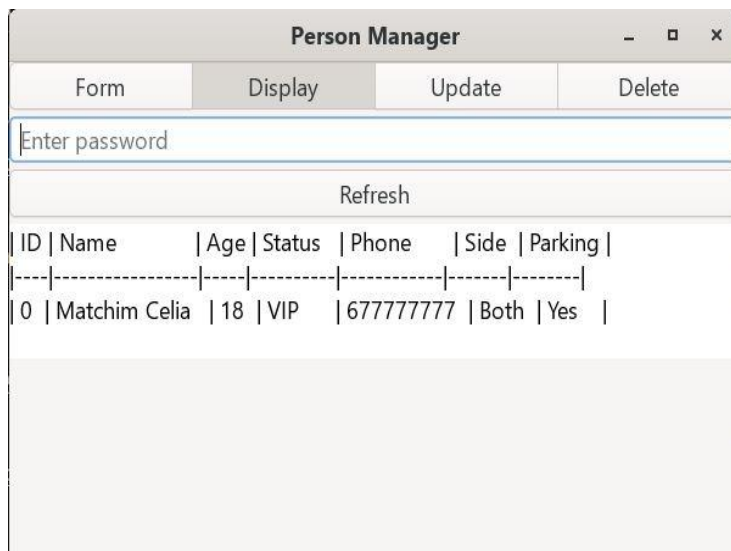
15.1.4 Executable File



person.c	24/03/2026 20:15	C Source File	22 Ko
person.exe	01/04/2026 21:23	Application	148 Ko

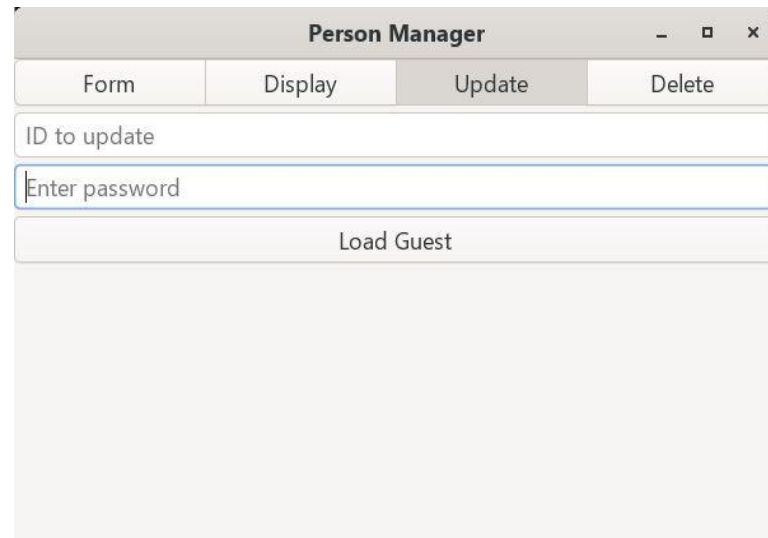
- Tests were carried out to verify that the Person management form performs its job correctly

(a) Person registration



ID	Name	Age	Status	Phone	Side	Parking
0	Matchim Celia	18	VIP	677777777	Both	Yes

(b) Person update



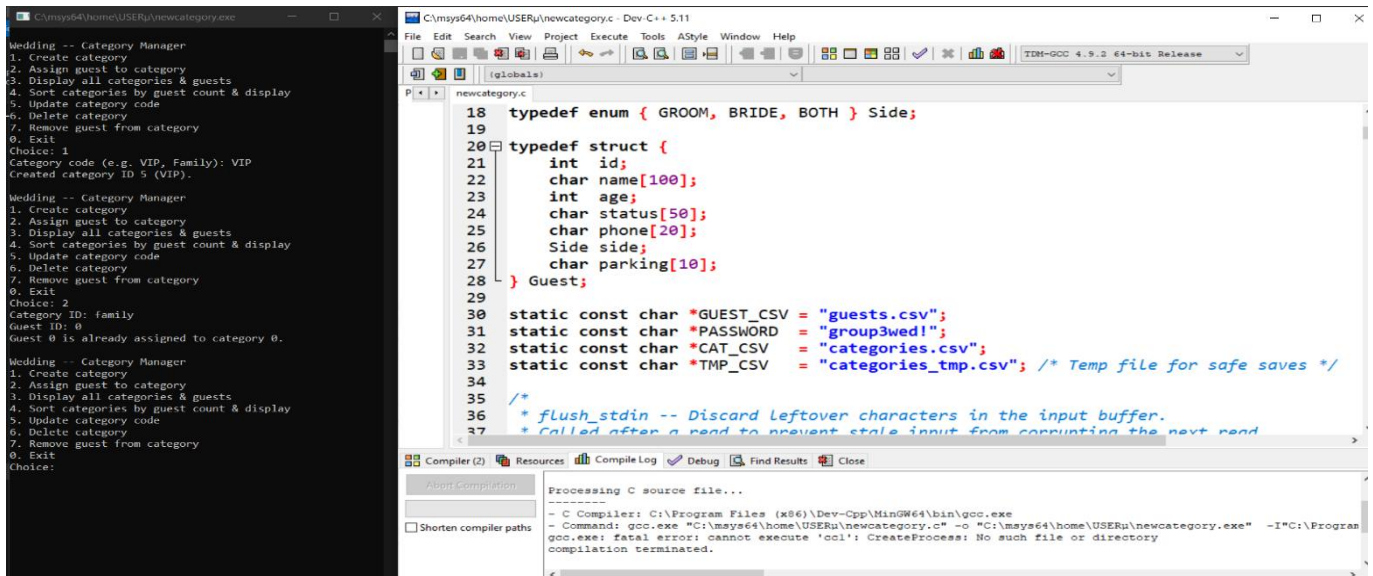
15.2 Category Management

The category management module was implemented in order to allow for creation of different categories at a wedding ceremony.

- Just like the person management module, alongside all the operations (Display, Update, Delete, Assign Guests)
- For assigning guests, the code was linked to the person management code, through the ID
- The code was thoroughly debugged using dev c++ software

15.2.1 Extract of Category Management C-code

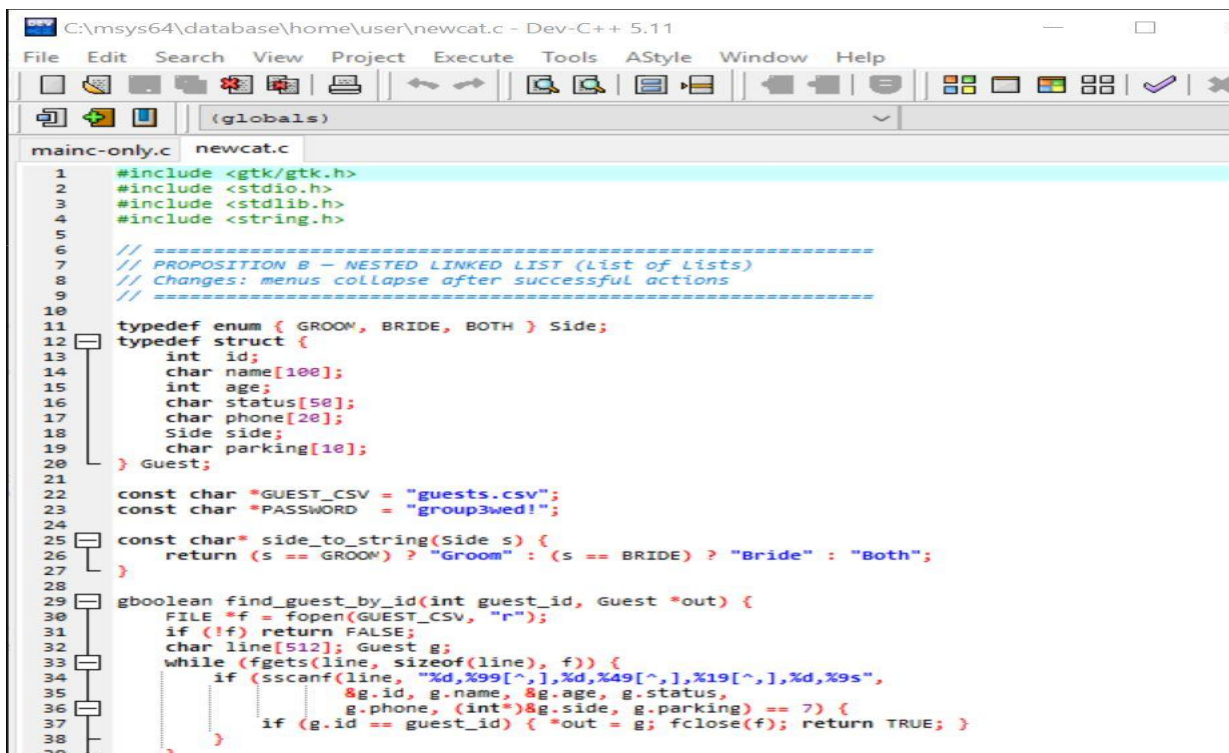
- Once the category management c program was correctly implemented, the GTK form version was created.



The image shows two windows from a Dev-C++ environment. The left window, titled 'C:\msys64\home\USERu\newcategory.exe', displays the runtime output of the program. It shows a menu for 'Wedding -- Category Manager' with options to create, assign, display, sort, update, delete, or remove categories and guests. The user has chosen to create a category named 'VIP' with ID 5. The right window shows the source code for 'newcategory.c'. It defines a 'Side' enum (GROOM, BRIDE, BOTH) and a 'Guest' struct with fields for id, name, age, status, phone, side, and parking. It also defines constants for CSV files and a password. The code includes a comment about flushing stdin to discard leftover characters.

```
18 typedef enum { GROOM, BRIDE, BOTH } Side;
19
20 typedef struct {
21     int id;
22     char name[100];
23     int age;
24     char status[50];
25     char phone[20];
26     Side side;
27     char parking[10];
28 } Guest;
29
30 static const char *GUEST_CSV = "guests.csv";
31 static const char *PASSWORD = "group3wed!";
32 static const char *CAT_CSV = "categories.csv";
33 static const char *TMP_CSV = "categories_tmp.csv"; /* Temp file for safe saves */
34
35 /* flush_stdin -- Discard leftover characters in the input buffer.
36  * Called after a read to prevent stale input from corrupting the next read
37  */
```

15.2.2 Extract of Category Management GTK code



The image shows a snippet of C code for a GTK application, located in 'mainc-only.c'. It includes headers for GTK, stdio, stdlib, and string. It defines the same 'Side' enum and 'Guest' struct as seen in the previous code block. It also defines constants for CSV files and a password. The code includes a function 'side_to_string' that converts a 'Side' enum value to a string. It also includes a function 'gboolean find_guest_by_id' that searches for a guest by ID in a CSV file.

```
1 #include <gtk/gtk.h>
2 #include <stdio.h>
3 #include <stdlib.h>
4 #include <string.h>
5
6 // =====
7 // PROPOSITION B - NESTED LINKED LIST (List of Lists)
8 // Changes: menus collapse after successful actions
9 // =====
10
11 typedef enum { GROOM, BRIDE, BOTH } Side;
12 typedef struct {
13     int id;
14     char name[100];
15     int age;
16     char status[50];
17     char phone[20];
18     Side side;
19     char parking[10];
20 } Guest;
21
22 const char *GUEST_CSV = "guests.csv";
23 const char *PASSWORD = "group3wed!";
24
25 const char* side_to_string(Side s) {
26     return (s == GROOM) ? "Groom" : (s == BRIDE) ? "Bride" : "Both";
27 }
28
29 gboolean find_guest_by_id(int guest_id, Guest *out) {
30     FILE *f = fopen(GUEST_CSV, "r");
31     if (!f) return FALSE;
32     char line[512];
33     while (fgets(line, sizeof(line), f)) {
34         if (sscanf(line, "%d,%99[^\n],%d,%49[^\n],%19[^\n],%d,%9s",
35             &g.id, g.name, &g.age, g.status,
36             g.phone, (int*)&g.side, g.parking) == 7) {
37             if (g.id == guest_id) { *out = g; fclose(f); return TRUE; }
38         }
39     }
40 }
```

- Haven properly runs the code; the csv file was created automatically below the category management.

15.2.3 Category ran successfully

```
user@DESKTOP-S24HAFG UCRT64 ~  
$ gcc newcat.c -o newcat.exe $(pkg-config --cflags --libs gtk4)  
user@DESKTOP-S24HAFG UCRT64 ~  
$ |
```

15.2.4 Executable file

newcat	03/04/2026 09:42	C Source File	26 Ko
newcat	03/04/2026 09:42	Application	168 Ko

ren
perly
ited
csv

file, test was carried out, similarly to those in the person management, to ensure functionality

15.2.5. Assignment Page

Wedding Category Manager [Nested List]

New Category	Assign Guest	Display	Update	Delete
--------------	--------------	---------	--------	--------

? Close

Create a new Category

Code:

Create Category

Wedding ♦ Category Manager [Nested List]					
New Category	Assign Guest	Display	Update	Delete	
Password: ●●●●●●●●					
Show All Sort by Guest Count					
NESTED LINKED LIST Categories: 1 Total guests assigned: 1 (No fixed guest limit per category)					
[Category 5 VIP (1 guest(s))					
[GuestID 0] Matchim Celia Age: 18 VIP 677777777 Both Parking: Yes					

16. Combination of person and category management onto a single window

Haven made the person and category management fully-functional; it is necessary to combine them in a single window where a user has the possibility of choosing which to fill.

- The c-code, combinedmodules, was written, thoroughly de bugged and converted to a GTK form, and the csv file generated subsequently
- Main code with c only

16.1 Extract of combinedmodules.c

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4
5 /*
6  * Module descriptions
7  */
8 static void print_banner(void)
9 {
10     printf("\n");
11     printf("===== \n");
12     printf(" WEDDING GUEST SYSTEM - GROUP 3 \n");
13     printf(" Management Portal \n");
14     printf("===== \n");
15     printf("\n");
16 }
17
18 static void print_module_info(void)
19 {
20     printf(" MODULE 01 - Person Management \n");
21     printf(" Register, update and manage every wedding guest \n");
22     printf(" with full validation and CSV persistence. \n");
23     printf(" Features: \n");
24     printf("   - Add & save guests (CSV) \n");
25     printf("   - Live field validation \n");
26     printf("   - Update guest information \n");
27     printf("   - Password-protected delete \n");
28     printf("   - Auto-refresh display \n");
29     printf("\n");
30     printf(" MODULE 02 - Category Management \n");
31     printf(" Organise guests into nested linked-list \n");
32     printf(" categories such as VIP, Family, Friends. \n");
33     printf(" Features: \n");
34     printf("   - Create named categories \n");
35     printf("   - Assign guests (nested list) \n");
36     printf("   - Sort by guest count \n");
37     printf("   - Update & delete categories \n");
38     printf("   - Remove guests from category \n");
39     printf("\n");

```

```

MODULE 01 - Person Management
Register, update and manage every wedding guest
with full validation and CSV persistence.
Features:
  A Add & save guests (CSV)
  A Live field validation
  A Update guest information
  A Password-protected delete
  A Auto-refresh display

MODULE 02 - Category Management
Organise guests into nested linked-list
categories such as VIP, Family, Friends.
Features:
  A Create named categories
  A Assign guests (nested list)
  A Sort by guest count
  A Update & delete categories
  A Remove guests from category

PASSWORD PROTECTED - group3wed!
Wedding Guest System v1.0
-----

1. Open Module 01 - Person Management
2. Open Module 02 - Category Management
0. Exit

Choice:

```

16.2 Extract of Main code with Css

- Css was added to the GTK inorder to beautify the window

```
#include <gtk/gtk.h>
#include <stdlib.h>
#include <string.h>

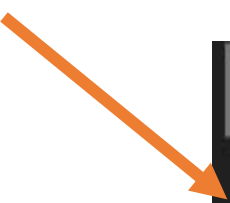
/*
-----
CSS - dark navy + gold palette
-----
*/
static const char *APP_CSS =
/* window */
"window {"
"  background-color: #0d1b2e;"
"}"

/* Top banner */
".banner {"
"  background: linear-gradient(135deg, #091422 0%, #0d1b2e 60%, #112038 100%);"
"  padding: 36px 48px 28px 48px;"
"  border-bottom: 1px solid rgba(201,168,76,0.25);"
"}"
".banner-eyebrow {"
"  color: #c9a84c;"
"  font-size: 18px;"
"  letter-spacing: 4px;"
"  font-weight: bold;"
"}"
".banner-title {"
"  color: #f5f0e8;"
"  font-size: 30px;"
"  font-weight: bold;"
"  margin-top: 6px;"
"}"
".banner-sub {"
"  color: rgba(245,240,232,0.45);"
"  font-size: 12px;"
"  margin-top: 4px;"
"  letter-spacing: 1px;"
"}"
".gold-rule {"
"  background: linear-gradient(90deg, transparent, #c9a84c, transparent);"
"  min-height: 1px;"
"  margin: 16px 0 0 0;"
}
```

16.3 combinemodules ran Successfully

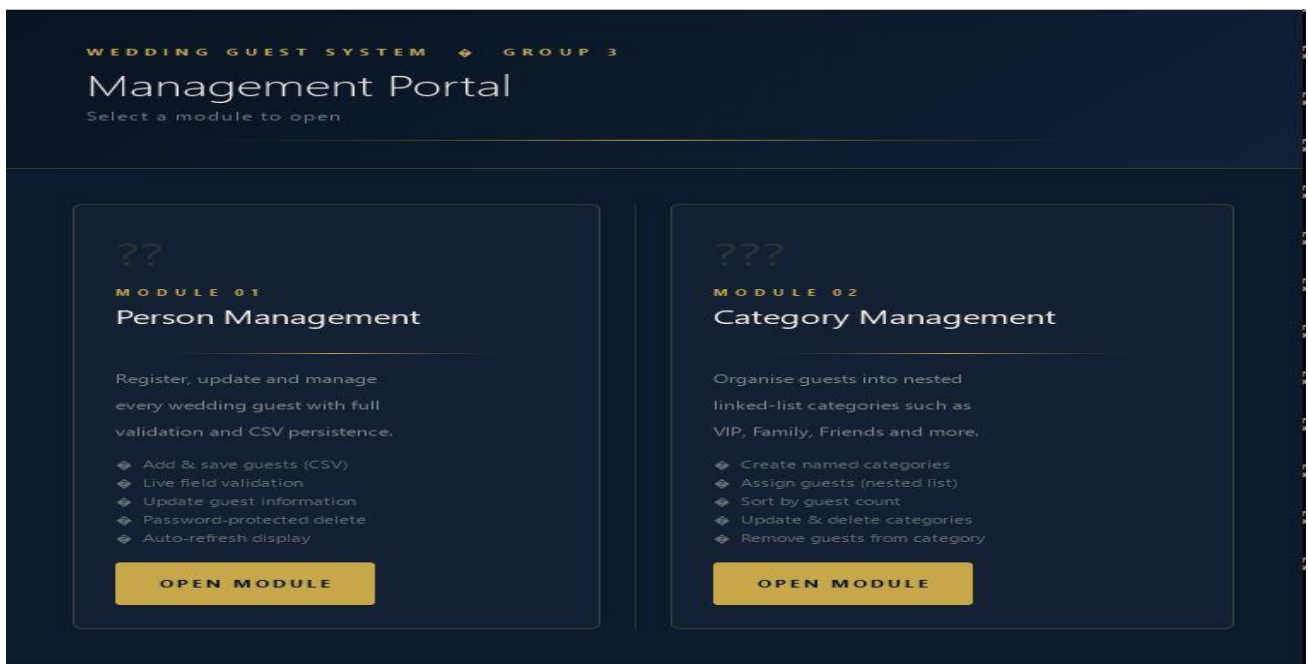
```
user@DESKTOP-S24HAFG UCRT64 ~
$ gcc mainapp.c -o mainapp.exe $(pkg-config --cflags --libs gtk4)
mainapp.c: In function 'on_launch_person':
mainapp.c:160:9: warning: 'gtk_message_dialog_new' is deprecated [-Wdeprecated-decl
160 |         GtkWidget *dlg = gtk_message_dialog_new(
    |         ^~~~~~
In file included from C:/msys64/database/ucrt64/include/gtk-4.0/gtk/gtk.h:194,
    from mainapp.c:16:
C:/msys64/database/ucrt64/include/gtk-4.0/gtk/deprecated/gtkmessagedialog.h:81:12: n
ote: declared here
  81 | GtkWidget* gtk_message_dialog_new      (GtkWindow      *parent,
    |         ^~~~~~
mainapp.c: In function 'on_launch_category':
mainapp.c:181:9: warning: 'gtk_message_dialog_new' is deprecated [-Wdeprecated-decl
181 |         GtkWidget *dlg = gtk_message_dialog_new(
    |         ^~~~~~
C:/msys64/database/ucrt64/include/gtk-4.0/gtk/deprecated/gtkmessagedialog.h:81:12: n
ote: declared here
  81 | GtkWidget* gtk_message_dialog_new      (GtkWindow      *parent,
    |         ^~~~~~
user@DESKTOP-S24HAFG UCRT64 ~
$ |
```


16.4. Executable file



 mainapp	03/04/2026 10:33	C Source File	13 Ko
 mainapp	03/04/2026 10:34	Application	153 Ko

16.5. Combinedmodules Window



17. THE GIFT MANAGEMENT

The gift management module was implemented in order manage the possible gifts at a wedding ceremony.

- It helps the couple to know what their guests are planning to give or offer them.
- Furthermore, just like the person and category management module, it carries alongside all the operations;
 - Register
 - Display

- Delete
- update gifts from guests

17.1 Extract of Gift Management C-code

```

/* ACTION: Add / Register a gift
/* ===== */
void action_add_gift(void) {
    printf("\n== Register a Gift ==\n");

    /* Step 1 = verify guest name (up to 3 tries) */
    char guest_name[100];
    int guest_id = -1;
    char category[50] = "Unknown";
    int verified = 0;

    for (int attempt = 0; attempt < 3 && !verified; attempt++) {
        read_line("Enter your full name (as registered): ", guest_name, sizeof(guest_name));
        if (verify_guest_name(guest_name, &guest_id, category, sizeof(category))) {
            verified = 1;
        } else {
            printf("Name not found in the guest list. Please re-enter the correct name.\n");
        }
    }

    if (!verified) {
        printf("Could not verify your name. Gift registration cancelled.\n");
        return;
    }

    printf("Welcome, %s! (ID: %d | Category: %s)\n", guest_name, guest_id, category);

    /* Step 2 = show catalogue and let guest choose */

```

```

***** Wedding Gift Management *****
=====
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice: 2
==== Display All Gifts ====
Password: group3wed!
No gifts recorded yet.
=====
***** Wedding Gift Management *****
=====
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice:

```

```

***** Wedding Gift Management *****
=====
1. Register a gift
2. Display all gifts
3. Delete a gift
4. Update a gift
0. Exit
=====
Choice: 1
==== Register a Gift ====
Enter your full name (as registered): Brenda
Name not found in the guest list. Please re-enter the correct name.
Enter your full name (as registered):

```

- Once the gift management c program was correctly implemented, the GTK version was created

17.2 Extract of Gift Management GTK code

```
37 static const char *GUEST_CSV = "guests.csv";
38 static const char *CAT_CSV = "category.csv";
39 static const char *GIFT_CSV = "gifts.csv";
40 static const char *TMP_GIFT = "gifts_tmp.csv";
41 static const char *PASSWORD = "group3wed!";
42
43 typedef struct { int id; const char *name; double price_fcfa; } GiftItem;
44
45 static const GiftItem CATALOGUE[GIFT_ITEM_COUNT] = {
46     { 1, "House", 45000000.0 },
47     { 2, "Car", 8500000.0 },
48     { 3, "Pots Set", 35000.0 },
49     { 4, "Spoon Set", 8000.0 },
50     { 5, "Fork Set", 8000.0 },
51     { 6, "Dishes Set", 25000.0 },
52     { 7, "Glass Set", 15000.0 },
53     { 8, "Microwave", 55000.0 },
54     { 9, "Customized Photo Frame", 12000.0 },
55     { 10, "Travel Tickets (per person)", 350000.0 },
56     { 11, "Hotel Reservation (per night)", 120000.0 },
57     { 12, "Wine (bottle)", 18000.0 },
58     { 13, "Smoothing Iron", 22000.0 },
59     { 14, "Bedside Table", 45000.0 },
60     { 15, "Cushion Set", 20000.0 },
61     { 16, "Curtain Set", 30000.0 },
62     { 17, "Television (43\")", 180000.0 },
63     { 18, "Flower Pots", 7000.0 },
64     { 19, "Kettle", 15000.0 },

```

17.3 Gift management ran successfully

	gift_gtk03.c	04/05/2026 02:19	C Source File	44 Ko
	gift_management_group3.exe	04/05/2026 03:21	Application	192 Ko

17.4 Executable file

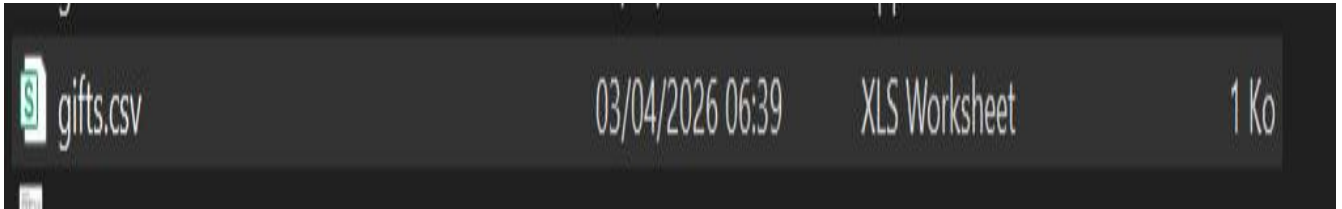
- Haven properly created the csv file, test was carried out, similarly to those of category and person management so as to insure functionality.
- - We then created the csv file in order to carry out the tests



```
USER@DESKTOP-KSHPOJE UCRT64 ~
$ gcc gift_gtk03.c -o gift_management_group3.exe $(pkg-config --cflags --libs gtk4) -lm

USER@DESKTOP-KSHPOJE UCRT64 ~
$ |
```

csv file



Register a gift

A screenshot of a web application titled 'Wedding Gift Manager - group3wed'. The application has four tabs: 'Register Gift', 'Display Gifts', 'Delete Gift', and 'Update Gift'. The 'Register Gift' tab is active. The form contains the following sections:

- Step 1 : Verify your identity (Full Name)**: A text input field with 'Matchim Celia' entered. Below it is a 'Verify Identity' button. The text 'Verified: Matchim Celia' is displayed below the button.
- Step 2 : Choose your gift**: A dropdown menu showing '1. House' with a value of '45000000 FCFA (68602.06 EUR)'. Below the dropdown, the unit price is '45000000 FCFA (68602.06 EUR)'. There is a quantity input field with '1' and minus/plus buttons. The total is '45000000 FCFA (68602.06 EUR)'.
- Step 3 : Thank-you name (optional)**: A text input field with the placeholder 'e.g. Aunt Marie (blank = use registered name)'. Below it is a 'Register Gift' button.

implemented registration of a gift by a guest using their name because they do not know the id generated when they registered.

- In order to register a gift, a guest must have first registered in the person module.
- The name Matchim Celia, previously registered in the person.c was successfully identified, showing the success of our code

Delete Gift

A screenshot of the 'Wedding Gift Manager - group3wed' web application with the 'Delete Gift' tab active. The form contains the following sections:

- Step 1 : Password**: A password input field with masked characters and a 'Confirm Password' button. The text 'Password accepted' is displayed below the button.
- Step 2 : Verify your identity (Full Name only)**: A text input field with 'Exactly as registered' entered. Below it is a 'Verify Identity' button.
- Step 3 : Gift Record to delete**: A text input field with 'Gift record ID (from Display tab)' entered. Below it is a 'Delete Gift Record' button.

Display Gift

Password accepted

Gift Item	Qty	Total FCFA	Total EUR
House	1	45000000	68602,06

Update Gift

Wedding Gift Manager - group3wed

Register Gift Display Gifts Delete Gift Update Gift

Step 1 : Password

.....

Password accepted

Step 2 : Verify your identity (Full Name only)

Full Name:

Record ID:

Step 3 : Select new gift and quantity (total auto-calculated)

1. House 45000000 FCFA (68602.06 EUR)

Unit price: 45000000 FCFA (68602.06 EUR)

Quantity:

Total: 45000000 FCFA (68602.06 EUR)

- To display, delete or update a gift, a password must be inserted without which these operations cannot be done.

NB:

- A password was used to prevent guests from accessing fields other than registration. Password: group3wed!

-For being lengthy, full codes could not be integrated in our report. Full codes are available in the GitHub repository at this link: <https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

- All the outputs of the codes are present in the repository too.

Combinedmodules C Window

The screenshot shows a C++ IDE with two windows. The left window displays the source code for `combinedmodules.c`, and the right window shows the program's execution output.

Source Code (combinedmodules.c):

```
217
218 26 predefined gift options that guests can choose from.
219 The last option (index 25) is "Cash Contribution" with price 0,
220 because the guest enters a custom amount.
221
222 static char *giftNames[] = {
223     "House", "Car", "Pot Set", "Spoon Set", "Fork Set",
224     "Dish Set", "Glass Set", "Microwave", "Photo Frame",
225     "Travel Ticket", "Hotel Reservation", "Wine", "Iron",
226     "Bedside Table", "Cushion Set", "Curtain Set", "Television",
227     "Flower Pot", "Kettle", "Cup Set", "Sheet Set", "Jewelry",
228     "Couple Watch", "Home Decor", "Gas Stove", "Cash Contribution"
229 };
230
231 static float giftPrices[] = {
232     1500000, 800000, 5000, 1000, 1000,
233     3000, 2000, 7000, 1500,
234     20000, 15000, 1000,
235     2500, 4000, 3000, 3500,
236     25000, 1000, 1500, 1200,
237     3000, 10000, 8000, 5000,
238     6000, 0
239 };
240
241 /*
242 FUNCTION: gift_show_menu()
243
244 PURPOSE:
245     Displays the full list of 26 gift options with their prices.
246     Called before a guest makes or updates their gift choice.
247
248 PARAMETERS: void
249 RETURN VALUE: void
250 */
251 static void gift_show_menu(void)
252 {
253     int i;
254     printf("\n--- Gift List ---\n");
255     for (i = 0; i < 26; i++) {
```

Execution Output:

```
? Multiple payment methods
? Password-protected display
? Update & delete gift records

PASSWORD PROTECTED - group3wed!
Wedding Guest System v1.0
-----

1. Open Module 01 - Person Management
2. Open Module 02 - Category Management
3. Open Module 03 - Gift Management
0. Exit

Choice: 3

=====
MODULE 03 - Gift Management
=====

1. Add a gift
2. Display all gifts
3. Display one gift
4. Update a gift
5. Delete a gift
0. Return to Main Menu

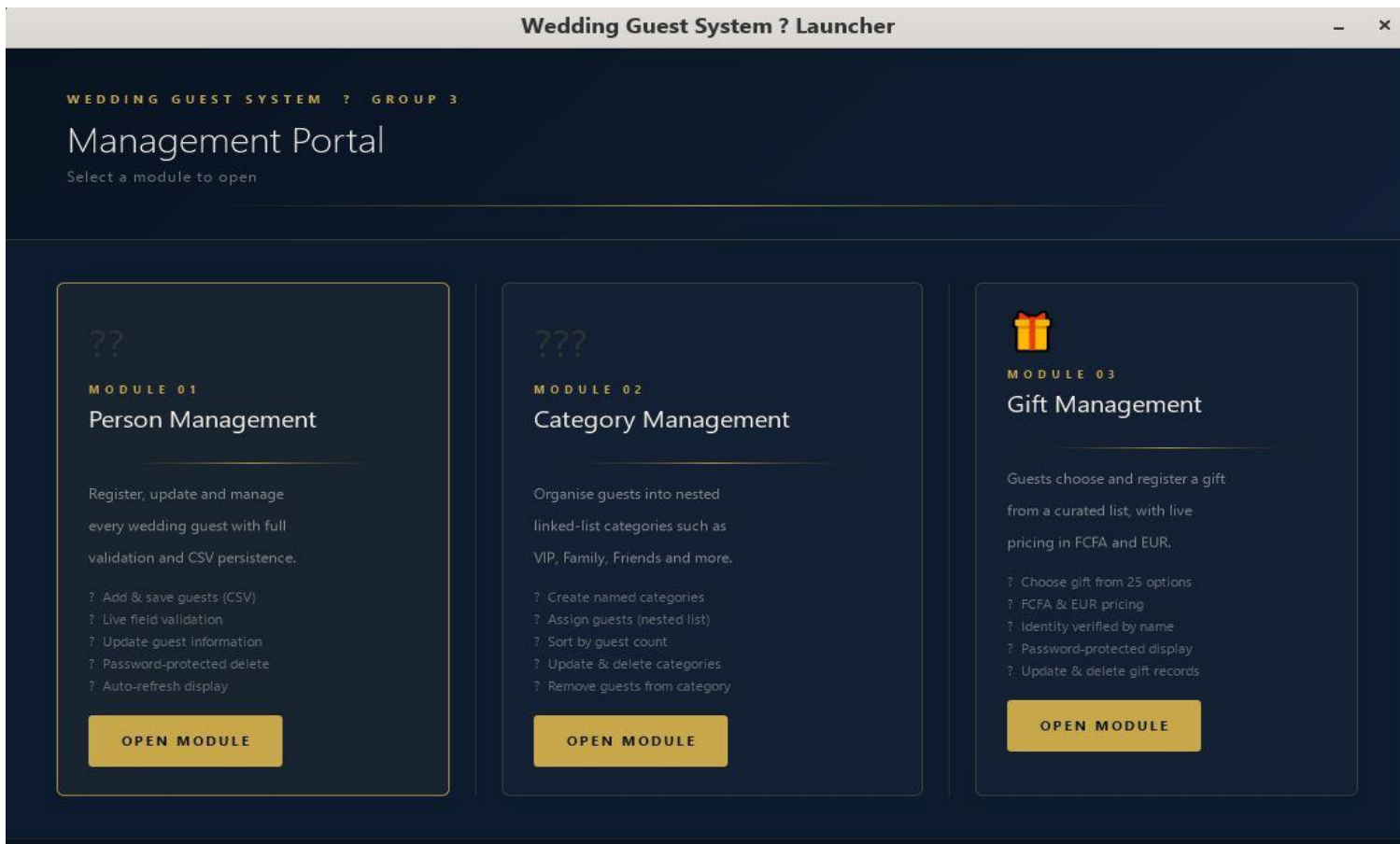
Choice: 2
Enter password: group3wed!
No gift records found.

=====
MODULE 03 - Gift Management
=====

1. Add a gift
2. Display all gifts
3. Display one gift
4. Update a gift
5. Delete a gift
0. Return to Main Menu

Choice:
```

✓ We then added the gift management to our combinedmodules.c



Combinedmodules GTK Window

18. CREATION OF MAKE FILE

A Makefile is a special text file used with a build automation tool called Make. It tells the tool how to compile, build, and manage a project.

- To better manage our program, we will build a Makefile in order to correlate all the modules.
- Create a file you name Makefile in the same folder as your c programs. You can create it as word, excel, blocnotes etc

Makefile.xlsx	04/04/2026 12:50	XLSX Worksheet	0 Ko
marriage.c	25/02/2026 01:24	C Source File	6 Ko

- Open your msys2MINGW64 terminal and type in the command “pacman -S make” to install make for windows

```

user@DESKTOP-S24HAFG MINGW64 ~
$ pacman -S make
resolving dependencies...
looking for conflicting packages...

Packages (1) make-4.4.1-2

Total Download Size:    0.50 MiB
Total Installed Size:  1.61 MiB

:: Proceed with installation? [Y/n]
:: Retrieving packages...
  make-4.4.1-2-x86_64    507.6 KiB   77.6 KiB/s   00:07 [#####] 100%
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files     [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) installing make          [#####] 100%
:: Running post-transaction hooks...
(1/1) Updating the info directory file...

user@DESKTOP-S24HAFG MINGW64 ~
$

```

- Next, verify the make version to assure that make was effectively

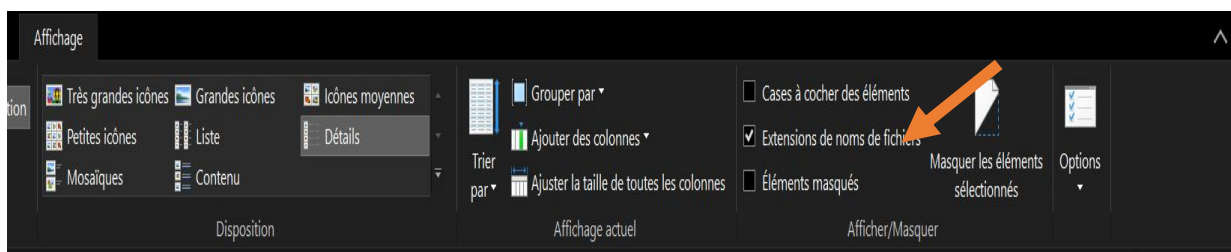
```

user@DESKTOP-S24HAFG MINGW64 ~
$ make --version
GNU Make 4.4.1
Built for x86_64-pc-msys
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

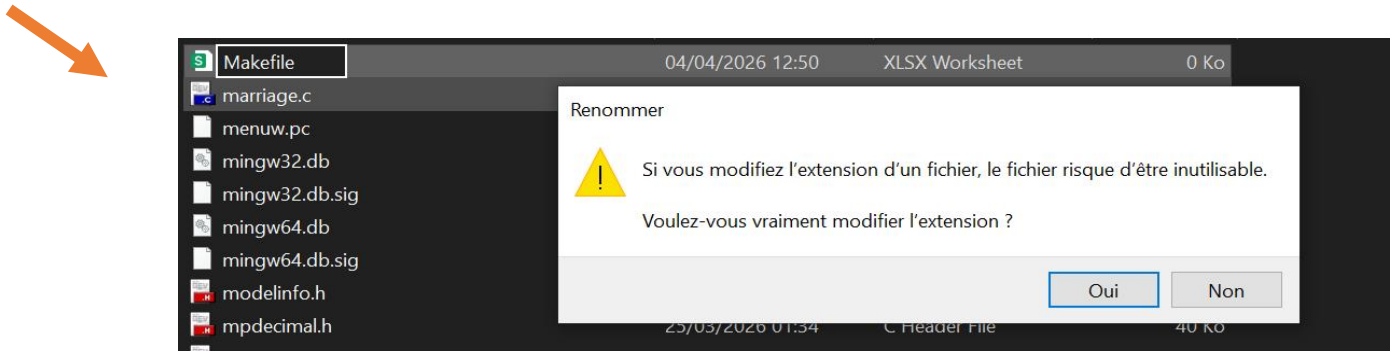
user@DESKTOP-S24HAFG MINGW64 ~
$ |

```

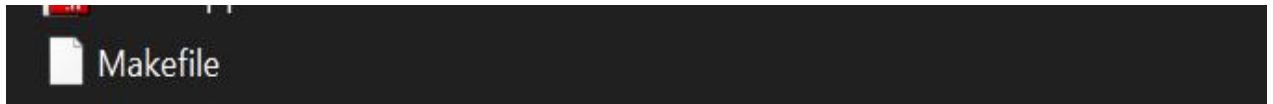
- show the extension of the file by going to the “Affichage” component and check the box “show ext and safeguard changes, so that your make file has no extension.



- Modify the extension by clicking “Oui” or “Yes.”



- Your Makefile has been created, you can now proceed with writing information into it.



- Open the MSYS2UCRT64 terminal and type in the command to search for the path where your Makefile is found

```
user@DESKTOP-S24HAFG UCRT64 ~
$ cd C:/msys64/database/home/user

user@DESKTOP-S24HAFG UCRT64 /c/msys64/database/home/user
$ |
```

- Type “make” in the terminal, and wait until all the different modules are all compiled.

```
[OK] launcher compiled

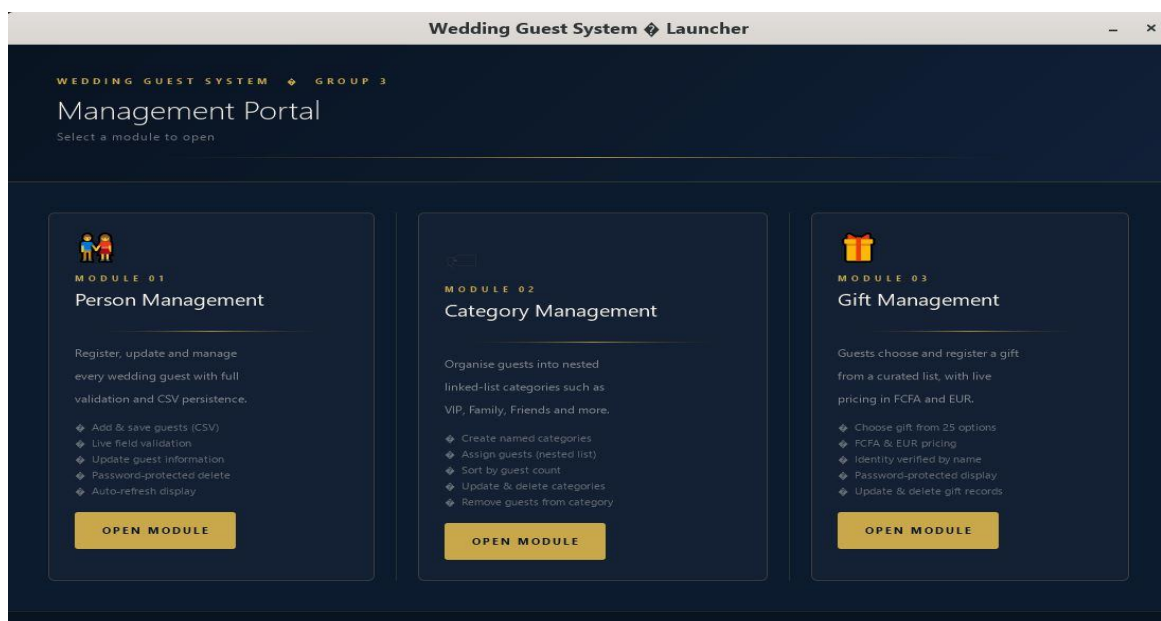
=====
All targets built successfully!
=====

user@DESKTOP-S24HAFG UCRT64 ~
$ |
```

- After your Makefile ran correctly, you can now proceed to launch the interface created with all your modules by typing the command “./launcher” in your terminal

```
user@DESKTOP-S24HAFG UCRT64 ~  
$ ./launcher  
  
(launcher.exe:5512): GLib-WARNING **: 08:32:21.092: Failed to determine console output code page: Le système n'a pas trouvé l'option d'environnement spécifiée.. Falling back to UTF-8  
  
(launcher.exe:5512): Pango-WARNING **: 08:32:21.130: Invalid UTF-8 string passed to pango_layout_set_text()  
  
(launcher.exe:5512): Pango-WARNING **: 08:32:21.149: Invalid UTF-8 string passed to pango_layout_set_text()
```

- Once it is done, a cube shaped GTK icon appears in your task bar. Click on it, you will see the interface with all three modules.



- You use the make clean prompt in order to clear away any actions done in the modules

```
user@DESKTOP-S24HAFG UCRT64 ~  
$ make clean  
rm -f person category gift combinedmodules \  
    gtkperson gtkcategory gift_gtk03 launcher \  
    temp.csv temp_renum.csv category_tmp.csv gifts_tmp.csv  
[OK] clean done
```

Optimizations Made

- We rendered the id of a person dynamic that is, if someone registers as a guest and has ID 0, and later on with draws, the ID of the persons following will immediately be adjusted; i.e the next will go to 0, and the following to 1...
- ❖ Github link: <https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>

19. CONCLUSION

With Modules 1,2 and 3 implemented and optimized, we are ready for the implementation of module 4 and 5, parking management and priority management.

20. PARKING MANAGEMENT IMPLEMENTATION

This module is optional for guests, as it applies only for guests who have registered a “Yes” in parking during guest registration in Person Management Module.

20.1 Pure C parking

It consists of two panels, the Administrator Panel and the Guests Panel.

```
-----
Parking Management System
-----
1. Guest Panel
2. Admin Panel
0. Exit
-----
Choice:
```

20.1.1 The Administrator Panel

```
Admin password: group3wed!
[OK] Access granted.

-----
ADMIN MENU
Config: not set yet
-----
1. Configure spots
2. View spot status
3. Cancel any booking by ID
4. Back
-----
Choice: 1
Number of parking spots (1-200): 5
Maximum hours per booking: 3
[ERROR] Cannot write to parking_spot.csv
[OK] Saved: 5 spots, max 3 hour(s).

-----
ADMIN MENU
Config: 5 spots | max 3 hour(s) per booking
-----
1. Configure spots
2. View spot status
3. Cancel any booking by ID
4. Back
-----
Choice: 2
-----
SPOT      STATUS      GUEST      TIME WINDOW
-----
P1        AVAILABLE
P2        AVAILABLE
P3        AVAILABLE
P4        AVAILABLE
P5        AVAILABLE
-----
Free: 5    |    Occupied: 0    |    Total: 5
-----

-----
ADMIN MENU
Config: 5 spots | max 3 hour(s) per booking
-----
1. Configure spots
2. View spot status
3. Cancel any booking by ID
4. Back
-----
```

This panel is accessed only by the manager of the marriage, and can only be accessed by mean of a code. It can perform the following operations:

- Define the number of parking spots for the Marriage ceremony
- Define the maximum number of parking hours per guest
- It adjusts itself when a parking spot is free.

20.2.1 The Guest Panel

```
-----
PARKING MANAGEMENT SYSTEM
-----

-----
MAIN MENU
-----
1. Admin panel
2. Guest panel
3. Exit
-----
Choice: 2
Enter your full name: Lesly Kemmogne

Welcome, Lesly Kemmogne!

-----
GUEST MENU   ÔÇô   Lesly Kemmogne
-----
1. Book a parking slot
2. View my bookings
3. Cancel a booking
4. Back
-----
Choice:
```

This panel was designed for the registration of guests, and can be accessed by these latter without the use of a password. It performs the following operations:

- Assigns a guest to a parking based on its availability
- Tells the guest the parking they have been assigned to
- Ask the guest for the parking he would like, and the time he will spend on that parking spot

Pure c parking assignment

GUEST MENU ÔÇô Lesly Kemmogne

- 1. Book a parking slot
- 2. View my bookings
- 3. Cancel a booking
- 4. Back

Choice: 1

Start hour (0-23): 14

Start minute (0-59): 34

Duration (hours): 1

BOOKING SUMMARY

Spot : P1

From : 14:34

To : 15:34

Duration : 1 hour(s)

Confirm booking? (y/n): y

[OK] Booking confirmed! Spot P1 reserved from 14:34 to 15:34.

20.2 GUI parking management

Parking Management System

P Parking Management System

Select a panel to continue



Guest Panel



Admin Panel

Assigning a guest to a parking

Welcome, **Lesly Kemmogne** ✓

Request a Parking Slot

Start Hour (0–23):

Start Minute (0–59):

Duration (hours):

- A guest can also search their name to view booked spot if they have forgotten.

Your Parking Summary

Spot : 1

From : 09:30

To : 10:30

Duration : 1 hour(s)

☒ Confirm Booking

Guest Parking

Enter your registered name to access parking booking.

Guest Parking

Enter your registered name to access parking booking.

Full Name:

[← Home](#)

Welcome, **Kounga Junior** ✓

Request a Parking Slot

Start Hour (0–23):

Start Minute (0–59):

Duration (hours):

Your Parking Summary

Spot : 1

From : 07:15

To : 08:15

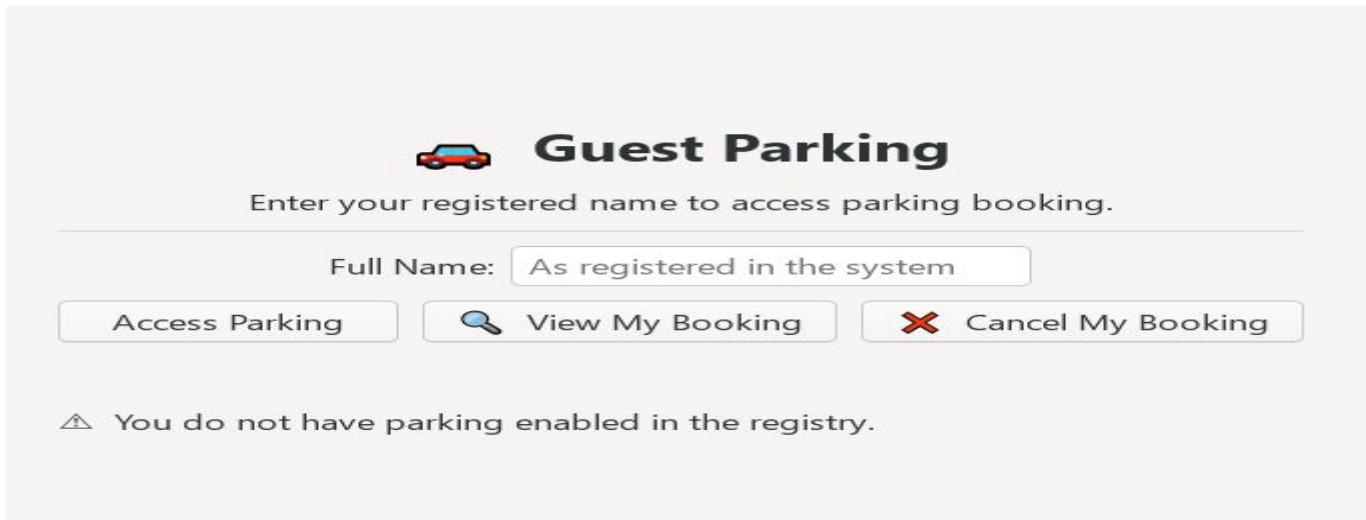
Duration : 1 hour(s)

☒ Confirm Booking

- Once a guest is assigned, they appear in parking_booking.csv

id	spot	Name	start_hr	mins	duration	end_hr	mins
1	1	Lesly Kemmogne	9	30	1	10	30
2	1	Kounga Junior	7	15	1	8	15

- When a guest registers “Yes” for parking and later on changes their mind and selects “No”, their booking is automatically cancelled and spot can be reallocated. Or, the guest can simply cancel their booking in the



Guest Parking

Enter your registered name to access parking booking.

Full Name:

⚠ You do not have parking enabled in the registry.

Example

If we change the parking status of Kounga Junior in person module from “Yes” to “No”, their booking is cancelled automatically.



Person Manager

Form | Display | **Update** | Delete

4

.....

Load Guest

Kounga Junior

23

Married

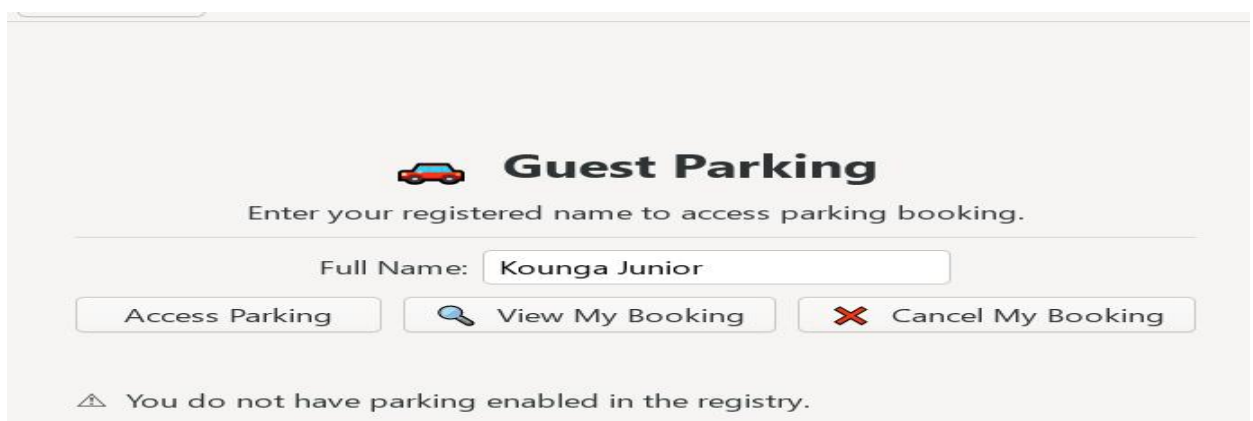
675854043

koungajunior@gmail.com

No

Side ☐ Groom ☒ Bride ☐ Both

Update



Guest Parking

Enter your registered name to access parking booking.

Full Name:

⚠ You do not have parking enabled in the registry.

- The spot was liberated and the booking was automatically cancelled in parking_booking.csv

Welcome, Lesly Kemmogne ✓

Request a Parking Slot

Start Hour (0-23): 9

Start Minute (0-59): 30

Duration (hours): 2

Free: 10 Occupied: 0

P1
AVAILABLE
Ready to book

P2
AVAILABLE
Ready to book

P3
AVAILABLE
Ready to book

P4
AVAILABLE
Ready to book

P5
AVAILABLE
Ready to book

P6
AVAILABLE
Ready to book

P7
AVAILABLE
Ready to book

P8
AVAILABLE
Ready to book

P9
AVAILABLE
Ready to book

P10
AVAILABLE
Ready to book

Preferred Spot No.: 2

Your Parking Summary

☒ Spot : 2

☐ From : 09:30

☐ To : 11:30

☐ Duration : 2 hour(s)

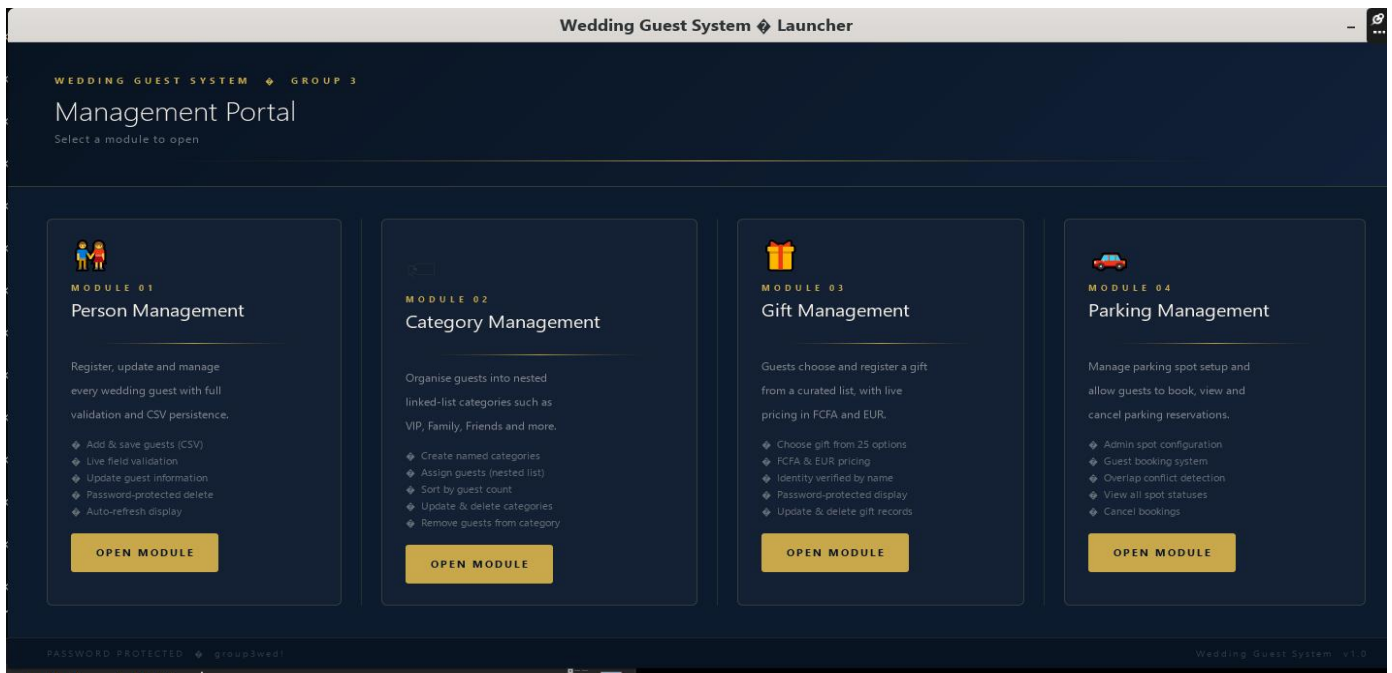
A	B	C
1,1,Lesly Kemmogne,9,30,1,10,30		

NB: This module has been designed in such a way that different guests can occupy the same parking spot, but not at the same time, as seen in the csv file

- The password to access the Administrator panel is “group3wed!”
- Guest can have only one booking
- For a guest to book a parking slot, they should be registered in persons.csv
- If a guest changes their mind, and select “No” for parking option in person module, their reserved parking spot is automatically cancelled, and can be attributed to other guests.

- Guest can select their parking spots themselves

21. ADDITION OF PARKING MODULE TO MAKEFILE



22. OPTIMIZATION OF PERSON & GIFT MANAGEMENT MODULE

The person management module was modified so that it asks for person's email, which will subsequently be used to send a guest their invitation:

22.1 Person Modification

Person Manager

Form

Display

Update

Delete

Full Name

Full Name

Age

17

Status

Single / Married ...

Phone

650123456

Email

guest@example....

Parking

No ▾

Side

☒ Groom ☐ Bride ☐ Both

Save

Person Manager

Form

Display

Update

Delete

group3wed

Refresh

ID	Name	Age	Status	Phone	Side	Parking	Email
0	Talla Daniella	16	VIP	674836353	Bride	No	Talladani@gmail.com
1	Matchim Celia	18	VIP	674839363	Bride	No	dzuyimmatchim@gmail.com
2	Lesly Yimdjo	22	Single	697854938	Groom	No	leslyyimdjo@gmail.com

22.2 Gift Optimization

Following the careful study of the actual gift management module, the following enhancements were carried out;

- Two panels were created; The Admins Panel and the Guest Panel

Wedding Gift Manager

group3wed

Please select your access level:

Guest Panel

Offer gifts & go to shop
(no password needed)

Admin Panel

Display / Delete / Update gifts
(password required)

- The Admins Panel can only be accessed by means of a password, and enables the manager of the marriage to;

Create gifts to add to the already-defined list of gifts

✓ View the already-existing gifts a guest has offered

Display Gifts
Delete Gift
Update Gift
Manage Catalogue

Current Gift Catalogue

ID	Gift Item	Price (FCFA)	Price (EUR)
1	House	45000000	68602,06
2	Car	8500000	12958,17
3	Pots Set	35000	53,36
4	Spoon Set	8000	12,20
5	Fork Set	8000	12,20
6	Dishes Set	25000	38,11
7	Glass Set	15000	22,87
8	Microwave	55000	82,85

Add a New Gift to the Catalogue

The new item will appear immediately in all guest / admin dropdowns.

Gift Name:

Price (FCFA):

Wedding Gift Manager - group3wed

Admin Panel Login
Enter the admin password to continue:

- The guest panel enables a guest to register a gift either alone or as a group with other guests, or simply make a financial contribution
- Each guest can offer the same gift only once.

One Guest → Gift(s)
Group Gift (Many Guests → One Gift)

1. House
45000000 FCFA (68602.06 EUR)

Unit price: 45000000 FCFA (68602.06 EUR)

Quantity:

Step 3 : Thank-you name (optional)

Your Gift Cart

(no gifts in cart yet)

Back
Guest Panel - Register a Gift

One Guest → Gift(s)
Group Gift (Many Guests → One Gift)

Register one gift offered jointly by multiple guests (minimum 2). All names must be verified before submitting.

Step 1 : Enter co-givers' names (minimum 2)

Guest 1:

Guest 2:

Step 2 : Choose the gift

1. House
45000000 FCFA (68602.06 EUR)

Unit price: 45000000 FCFA (68602.06 EUR)

Quantity:

One Guest → Gift(s) Group Gift (Many Guests → One Gift) **Financial Contribution**

Prefer to contribute financially instead of a physical gift? Fill in your details below.

Step 1 : Verify your identity (Full Name)

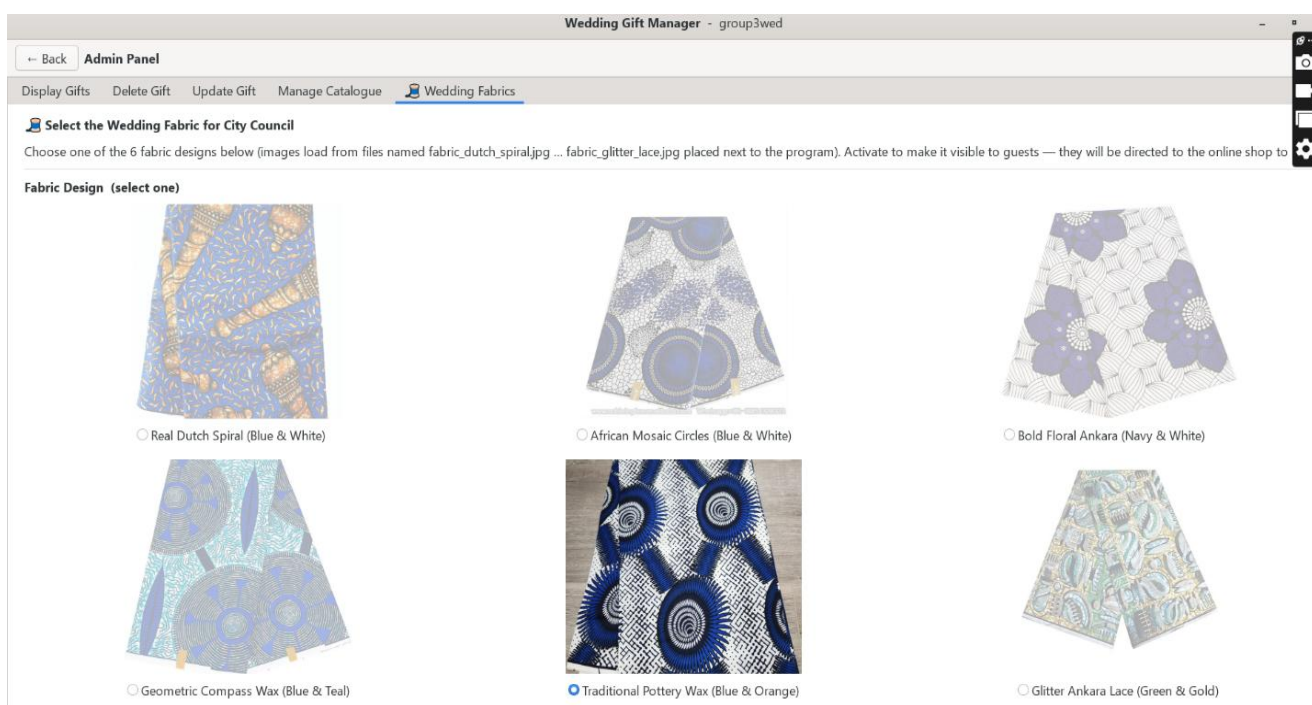
Full Name:

Step 2 : Contribution Details

Amount (FCFA):

Payment Method: ☒  Mobile Money ☐  Credit Card

- A section has been added in order for the guest to buy the dress of the wedding or not. If the former is chosen, they will be directed to the appropriate website
- When administrator selects a dress among a variety of choices, it appears to guest in guest panel, they can change whenever they want, and it automatically changes in guest panel.



- Once clicked, they are directed to the website selling it.

[← Back](#) **Guest Panel** – [Register a Gift](#)

[One Guest → Gift\(s\)](#) [Group Gift \(Many Guests → One Gift\)](#) [Financial Contribution](#) [Wedding Fabric](#)

Selected Fabric
Traditional Pottery Wax (Blue & Orange)



✓ Available — 1 interest registered. Click below to go to the online shop and purchase.

NB: For the images to effectively appear, they should be located in the same folder as the Graphical User Interface that is the msys2 folder

23. CREATION OF INVITATIONS SYSTEM

Following the creation and implementation of all the various modules, it is now time to generate the invitations for the guests;

- The manager can generate templates for the invitations, as seen below, with different colour possibilities

Create

Display

Send E-mail

Delete

Husband's Name:

Henry

Son of:

Mr & Mrs Nkambe

Wife's Name:

Daniella

Daughter of:

Mr & Mrs Edoh

Wedding Venue:

Hilton Hotel, Yaounde

Wedding Date:

25 December 2026

Dress Code:

Blue and Gold

Suggestions: Blue and Gold, Black and Pink,

— Event Times —

City Council Time:

09:00AM

Church Time:

11:00AM

Reception Time:

06:00PM

Colour Theme:

Royal Gold

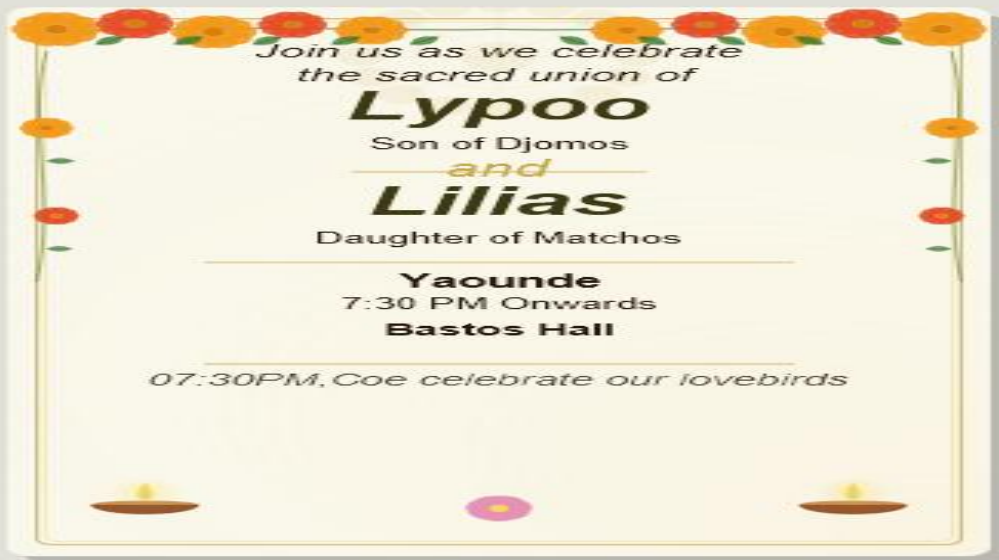
Personal Message:

We hope to have you with us!

Save Invitation

Invitation Preview

- The preview appears in the display;



Invitation #2



- Each guest can receive their invitations by email
- To configure sending of invitation by emails
 - Open the msys2MINGW64 terminal and enter the Command “pacman -S mingw-w64-ucrt-x86_64-curl”

```
M ~
user@DESKTOP-S24HAFG MINGW64 ~
$ pacman -S mingw-w64-x86_64-curl
```

- Install all the packages


```

user@DESKTOP-S24HAFG MINGW64 ~
$ pacman -S mingw-w64-x86_64-curl
resolving dependencies...
looking for conflicting packages...

Packages (25) mingw-w64-x86_64-brotli-1.2.0-1  mingw-w64-x86_64-c-ares-1.34.6-1
mingw-w64-x86_64-ca-certificates-20250419-1
mingw-w64-x86_64-gcc-libs-16.1.0-2
mingw-w64-x86_64-gettext-runtime-1.0-1  mingw-w64-x86_64-gmp-6.3.0-2
mingw-w64-x86_64-gnutls-3.8.13-1  mingw-w64-x86_64-libffi-3.5.2-1
mingw-w64-x86_64-libiconv-1.19-1  mingw-w64-x86_64-libidn2-2.3.8-4
mingw-w64-x86_64-libpsl-0.21.5-3  mingw-w64-x86_64-libssh2-1.11.1-2
mingw-w64-x86_64-libtasn1-4.21.0-1
mingw-w64-x86_64-libunistring-1.4.2-1
mingw-w64-x86_64-libwinpthread-14.0.0.r37.g2bfe61fba-1
mingw-w64-x86_64-nettle-3.10.2-1  mingw-w64-x86_64-nghttp2-1.69.0-1
mingw-w64-x86_64-nghttp3-1.58.0-1  mingw-w64-x86_64-ngtcp2-1.22.1-1
mingw-w64-x86_64-openssl-3.0.12-1  mingw-w64-x86_64-p11-kit-0.26.2-1
mingw-w64-x86_64-tzdata-2026a-1  mingw-w64-x86_64-zlib-1.3.2-2
mingw-w64-x86_64-zstd-1.5.7-2  mingw-w64-x86_64-curl-8.20.0-1

Total Download Size: 19.90 MiB
Total Installed Size: 116.37 MiB

Proceed with installation? [Y/n]
Retrieving packages...
mingw-w64-x86_64... 849.4 KiB 76.7 KiB/s 00:11 [#####] 100%
mingw-w64-x86_64... 1650.2 KiB 139 KiB/s 00:12 [#####] 100%
mingw-w64-x86_64... 2.3 MiB 176 KiB/s 00:13 [#####] 100%
mingw-w64-x86_64... 728.5 KiB 332 KiB/s 00:02 [#####] 100%
mingw-w64-x86_64... 577.8 KiB 383 KiB/s 00:02 [#####] 100%
mingw-w64-x86_64... 466.4 KiB 563 KiB/s 00:01 [#####] 100%
mingw-w64-x86_64... 563.7 KiB 180 KiB/s 00:03 [#####] 100%
mingw-w64-x86_64... 417.3 KiB 424 KiB/s 00:01 [#####] 100%
mingw-w64-x86_64... 395.0 KiB 554 KiB/s 00:01 [#####] 100%
mingw-w64-x86_64... 314.1 KiB 529 KiB/s 00:01 [#####] 100%
mingw-w64-x86_64... 444.5 KiB 113 KiB/s 00:04 [#####] 100%
mingw-w64-x86_64... 283.0 KiB 159 KiB/s 00:02 [#####] 100%
mingw-w64-x86_64... 1128.3 KiB 42.5 KiB/s 00:27 [#####] 100%
mingw-w64-x86_64... 654.2 KiB 208 KiB/s 00:03 [#####] 100%
mingw-w64-x86_64... 311.2 KiB 87.7 KiB/s 00:04 [#####] 100%
mingw-w64-x86_64... 199.2 KiB 74.4 KiB/s 00:03 [#####] 100%
mingw-w64-x86_64... 221.0 KiB 114 KiB/s 00:02 [#####] 100%
mingw-w64-x86_64... 7.8 MiB 131 KiB/s 01:02 [#####] 100%
Total (18/25) 19.9 MiB 239 KiB/s 01:25 [#####] 100%

```

Enter the Invitation ID and your password to send.

- Click 'Search for Email' to browse guests and select a recipient.
- Or type an address directly in 'Recipient E-mail'.
- 'Send to All Guests': sends to every guest with an e-mail in persons.csv.

Invitation ID:

Recipient E-mail:

3 guest(s) with e-mail — click a row to fill the address field:

Talla Daniella	Talladani@gmail.com
Matchim Celia	dzuyimmatchim@gmail.com
Lesly Yimdjo	leslyyimdjo@gmail.com

Password:

24. CREATION OF LOGIN MENU OF ALL WIGMS MODULES

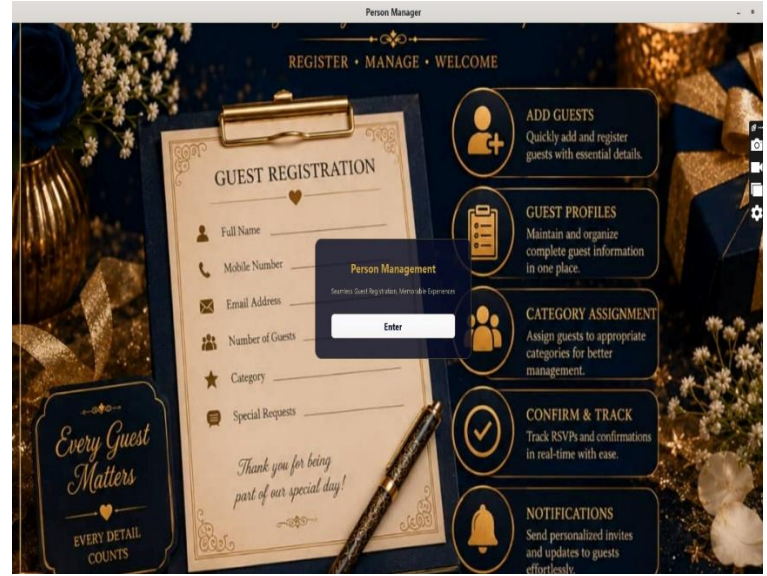
It is about the presentation of each module to the user. Here, every section of our WIGMS (Wedding Invitation and Gift Management System) needs to appear in a positive way to our user. In order to insure it, we used some images that will draw the attention and the interest of the user, give him some overview of what he will do in each section.

24.1. THE PERSON MANAGEMENT SECTION

As seen upstairs, it is the section where we can register a guest, set the guest profile based on his information entered, display all guest... The interface presenting this section is:

Image used

Login produced



- If you press enter, you will have access to all the features of the person management section listed up

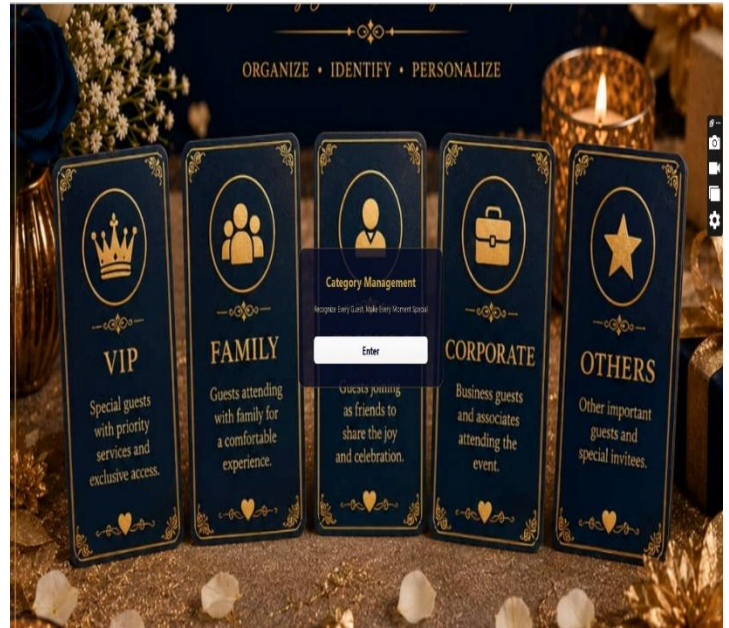
24.1 THE CATEGORY MANAGEMENT SECTION

As seen upstairs, it is the section where we can group guest depending on their category like: family members, friends... The interface presenting this section is:

Image used



Login Produced



If you press enter, you will have access to all the features of the category management section listed up

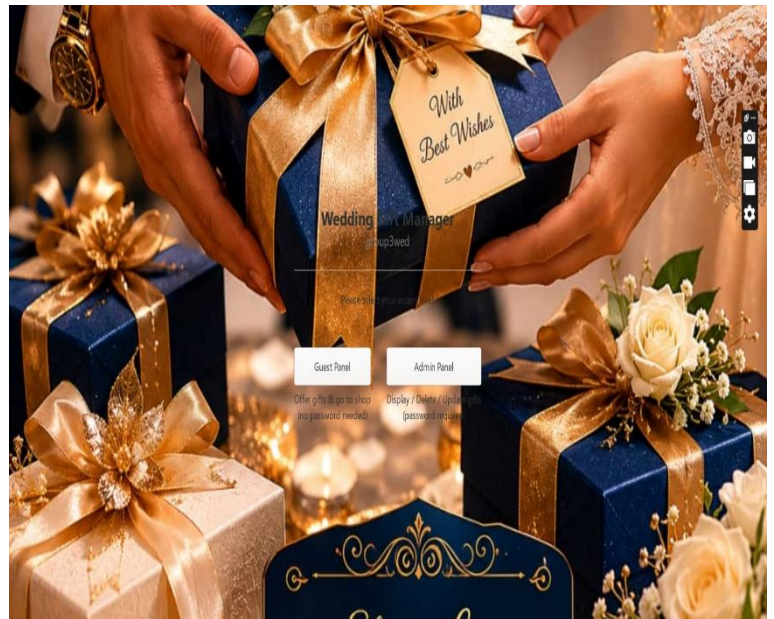
24.2 THE GIFT MANAGEMENT SECTION

As seen upstairs, it is the section which deals both with guest and the couple. Here, this section is used by the guest to offer a gift to the wedding couple. The couple uses this interface to see who offers them a gift... interface presenting this section is:

Image used



Login produced



- If you press enter, you will have access to all the features of the gift management section listed up.

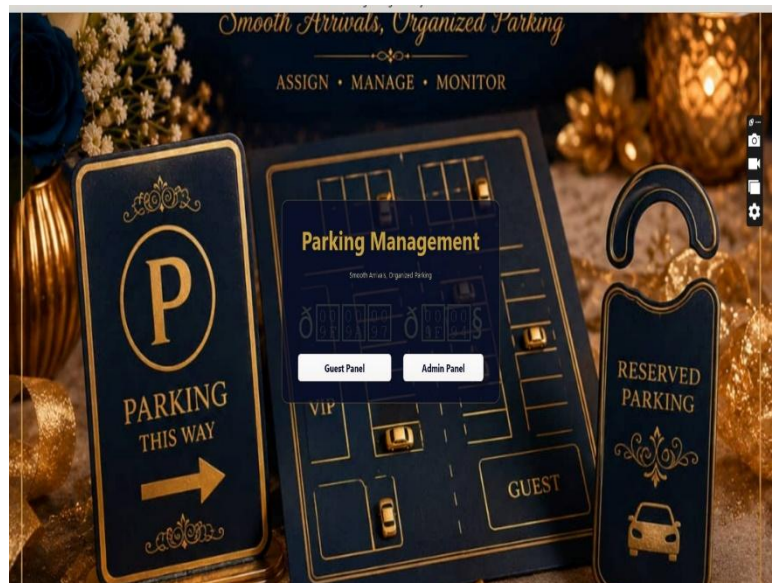
24.3 THE PARKING MANAGEMENT SECTION

As seen upstairs, it is the section where we can register a guest, set the guest profile based on his information entered, display all guest... The interface presenting this section is:

Image used



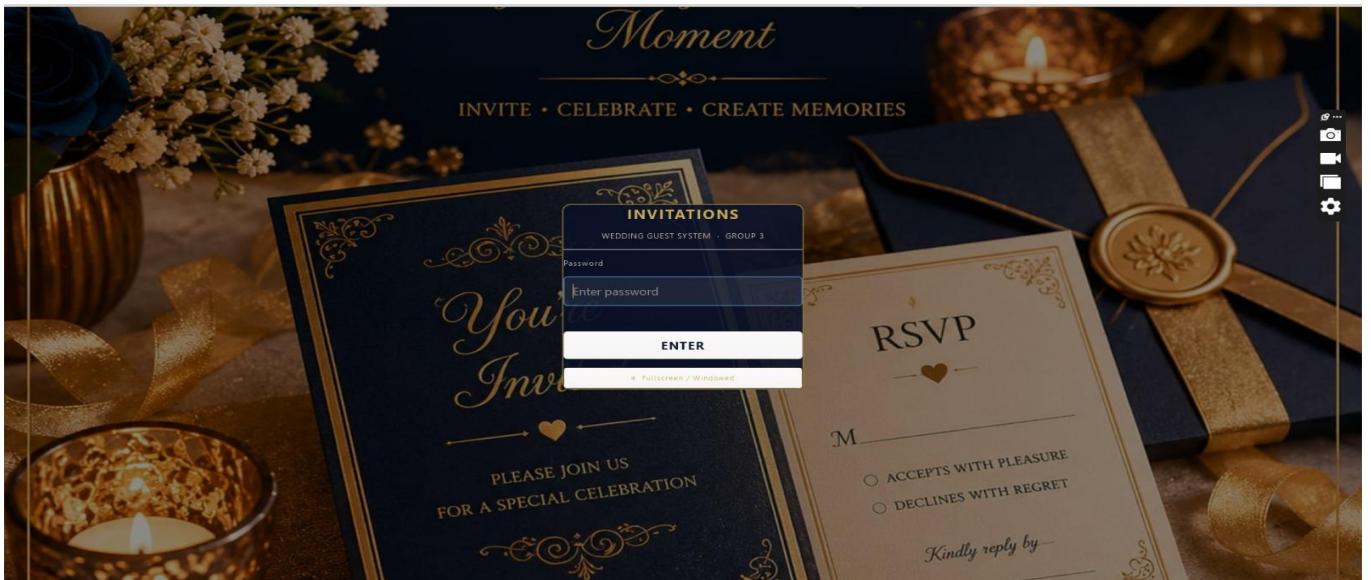
Interface produce



If you press enter, you will have access to all the features of the parking management section listed up.

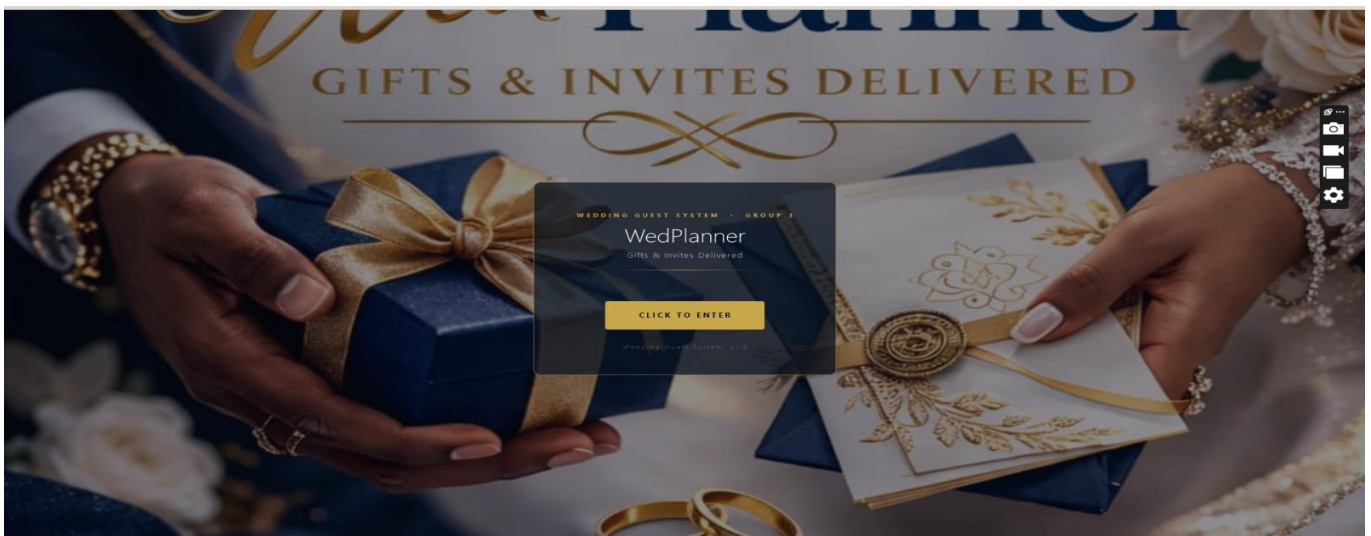
24.4 THE INVITATION SECTION

As seen upstairs, it is the section where the couple design their invitation card. Its presentation is:



24.5 THE MAKEFILE SECTION

As seen upstairs, it is the section where we combined all the modules. In fact, it is the first WIGMS presentation when being opened. The interface presenting this section is



NB

- All the different images must be found in the folder containing the gtk libraries

Module	Image Name
Person	person_bg.jpg
Category	category_bg.jpg
Gift	gift_bg.jpg
Parking	parking_bg.jpg
invitations	invitation_bg.jpg
makefile	mainwindow_bg.jpg

- All the images are found in the github under a folder called **Backgrounds**
- Github link: <https://github.com/celia-pixel-rgb/Project-Wedding-Ceremony>